

BRANCH SELECTION AND GLOBAL POLARIZATION IN THE CONNES–CCM ROUTE TO RH: FINITE PRIME–GAMMA INERTIA, TATE BOUNDARY GEOMETRY, AND THE DIVISOR-SENSITIVITY OBSTRUCTION

DAVID ALEJANDRO TREJO PIZZO

ABSTRACT. We organize the Connes–Consani–Moscovici and Suzuki operator programs as a single route from finite semilocal models to the global Weil criterion. The first part develops a joint prime–Gamma finite-source refinement. It keeps the ordinary von Mangoldt channels, the archimedean channel, the scalar threshold and the exact radical in one Hermitian system. Loewner monotonicity, Haynsworth additivity, scalar Schur innovations, adaptive Krylov recovery and a heat-core exhaustion remove the finite-selection, tail and form-core losses. An endpoint-corrected prolate ladder proves $|\mathcal{R}_L^+| \ll \lambda^7 d_4$, while a dyadic divisor estimate gives $\|b_L^+\| \ll \lambda^4 \sqrt{(1+L)d_4} \rightarrow 0$. The remaining sign in that realization is the strict physical source surplus over the anti-shorter deficit.

The second part constructs the local geometric data that this surplus was trying to express. A cyclic root Laplacian has pseudodeterminant N^2 ; the determinant increment from mp^{k-1} to mp^k is $\log p$, and its normalized root-stratum incidence is $p^{-k/2}$. Their product is exactly $\Lambda(p^k)/\sqrt{p^k}$. At the infinite place, a positive Gamma spin operator and a two-dimensional polar plane give, by zeta determinants, the complete factor $\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$. A shared Tate–Gamma boundary, matched Abel finite part and covariant charge mixing assemble these data without discarding prime–prime phases. All these constructions are source-defined and use neither zeros nor the Weil sign.

We then determine the precise limitation of the ordinary Hilbert completion. The complete charged pushout has absolutely continuous scale spectrum. An operator-valued dense-range theorem shows that its Gamma compliance and unitary charge mixing cannot create point spectrum. Consequently every equivariant map from the resonant CCM degree one into this completion annihilates the critical-line eigenclasses. Thus the local polarization is valid, but this global completion cannot be the missing polarization of CCM degree one.

Finally, we synthesize six terminal formulations reached by the route—a signed limit, a Schur surplus, a Rosati factorization, an equivariant Hilbert majorant, closure-faithful descent, and a positive global polarization. Each contains the complete Weil sign. Their common endpoint is analyzed by two complementary obstructions. Four forgetful functors are divisor-blind under explicit deformations, while the ordinary Hilbert pushout fails separately by point-versus-continuous spectral type. We formulate both mechanisms and isolate two open programs. One is geometric: construct an arithmetic intersection product and Hodge-index polarization on a suitable square of the arithmetic curve. The other is analytic but necessarily divisor-sensitive: construct a resonant nuclear or derived completion retaining evaluation jets before Hilbert reduction. No proof of either final positivity theorem is claimed.

CONTENTS

1.	Terminal Weil criterion	2
2.	The CCM finite spectral chain	3
3.	Localized operator realization	4

Independent Researcher. dtrejopizzo@gmail.com.

4. Finite-to-global branch convergence	4
5. Conditional closure of the CCM route	5
6. Localized realization and the global limit	5
7. Branch selection as a finite source theorem	6
8. Finite branch-selection refinement	7
8.1. Controlled exterior and cross residuals	8
9. Cofinal prime–Gamma inertia matching	13
9.1. Complete and finite joint heads	13
9.2. The strict finite frontier	14
9.3. Radical shorting and the inertia staircase	15
9.4. Geometry of a large-prime channel	16
9.5. Finite observability and the exact gain budget	16
9.6. Scalar innovations of the completed staircase	17
9.7. Local detection and cumulative adaptive gain	18
9.8. Radical-conditioned tail gain	19
9.9. Radical-conditioned Christoffel and finite determinant exhaustion	21
9.10. Signed amplitude transfer and adaptive Krylov recovery	23
9.11. Second-level spectral dispersion	24
9.12. Heat-core exhaustion of the radical complement	26
9.13. The compensated frontier and closure	27
10. Tate boundary geometry and the global polarization audit	30
10.1. Finite root geometry and the literal von Mangoldt mass	30
10.2. Poisson holonomy, heat lift and local unitary triviality	31
10.3. The Gamma spin and the polar determinant	32
10.4. Matched boundary cancellation and the full finite part	33
10.5. Covariant mixing of all prime-power charges	34
10.6. The global polarization target	35
10.7. The absolutely continuous completion no-go	35
10.8. Six terminal formulations and their common content	37
10.9. Two open programs	38
11. Conclusion	39
References	40

1. TERMINAL WEIL CRITERION

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \Xi(z) = \xi\left(\frac{1}{2} + iz\right).$$

For a test function f on $\mathbb{R}$, let $W(f)$ denote Weil's explicit formula distribution, normalized by

$$(1.1) \quad \begin{aligned} W(f) = & \int_{\mathbb{R}} f(x)(e^{x/2} + e^{-x/2}) dx - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} f(\log n) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} f(-\log n) \\ & - (\log 4\pi + \gamma) f(0) - \int_0^\infty \frac{f(x) + f(-x) - 2e^{-x/2} f(0)}{e^{x/2} - e^{-x/2}} dx. \end{aligned}$$

The associated Hermitian quadratic form is

$$(1.2) \quad \mathcal{Q}_W(v_1, v_2) = W(v_1 * \tilde{v}_2), \quad \tilde{v}(x) = \overline{v(-x)}.$$

Theorem 1.1 (Weil criterion [1, 2]). *The Riemann Hypothesis is equivalent to*

$$(1.3) \quad \mathcal{Q}_W(v, v) \geq 0 \quad (v \in C_c^\infty(\mathbb{R})).$$

Remark 1.2. The proof skeleton below is therefore not seeking a new final criterion. The final criterion is (1.3). Connes' route is a way of constructing (1.3) as the limit of finite or localized self-adjoint models.

Remark 1.3 (Attribution and novelty boundary). The terminal positivity criterion is Weil's criterion, in the noncommutative-geometric form developed by Connes. The Hilbert-space background around Sonin spaces, co-Poisson summation and radical-type quotients belongs to Burnol, Connes–Consani, Connes–Consani–Moscovici, Connes–van Suijlekom and Suzuki [12, 13, 3, 6, 4, 9, 11]. Related Connes–Consani developments around the arithmetic curve and absolute geometry, and Suzuki's Dirichlet- L variant, are part of the same operator-theoretic landscape but are not used as hidden inputs below [7, 8, 10]. The contribution of this paper starts where these inputs leave an unsigned branch-selection problem: finite-source Schur inertia, prime-by-prime innovation, observability staircases, adaptive Krylov recovery, the six-mode prolate residual estimates, and the heat/hybrid form-core exhaustion. Classical linear-algebraic devices used below, such as shorted operators, Haynsworth inertia additivity, Loewner order and total positivity, are cited at their first structural use.

2. THE CCM FINITE SPECTRAL CHAIN

Put $L = 2 \log \lambda$. The semilocal construction restricts the Weil form to a finite Fourier space

$$(2.1) \quad V_{L,N} = \text{span}\{e^{2\pi i k x/L} : |k| \leq N\}.$$

Let $W_{L,N}$ be the corresponding finite Weil matrix. It contains the polar, archimedean and prime-power terms of the semilocal trace formula. The central finite structural facts used by the proof skeleton are those of the CCM/Connes and Connes–van Suijlekom finite programs [6, 5, 4].

Definition 2.1 (Finite CCM package). For each L, N , the finite package consists of:

$$(2.2) \quad \varepsilon_{L,N} = \min \text{Spec } W_{L,N}, \quad T_{L,N} = W_{L,N} - \varepsilon_{L,N} I,$$

the radical line $\mathbb{C}\xi_{L,N} = \ker T_{L,N}$ when the bottom is simple, and the positive quotient

$$(2.3) \quad \mathcal{K}_{L,N} = (\mathbb{C}\xi_{L,N})^\perp, \quad S_{L,N} = T_{L,N}|_{\mathcal{K}_{L,N}}.$$

Connes–Consani–Moscovici identify the finite commutator structure, construct the quotient operator, and obtain self-adjoint finite spectral data after shifting and quotienting [6]. Connes' 2026 extremal construction adds the prolate model k_L , whose transform approximates Ξ . The component used in the closure theorem is the following finite spectral statement.

Theorem 2.2 (Finite critical-line CCM–Connes model [6, 5, 4]). *For each finite level where the radical quotient is defined, the quotient operator is self-adjoint in the positive quotient metric. Its finite spectral divisor is located on the critical line. Moreover there is an explicit prolate model k_L such that*

$$(2.4) \quad \widehat{k}_L(z) \longrightarrow \Xi(z)$$

locally uniformly on compact subsets of the open strip $|\operatorname{Im} z| < 1/2$.

Remark 2.3. Theorem 2.2 is the strong finite achievement: the finite models have real spectral parameters, and the model transform tends to the right entire function in the interior strip. What it does not by itself give is lower-semicontinuity of the indefinite Weil form in the global limit.

3. LOCALIZED OPERATOR REALIZATION

Suzuki replaces the distributional Weil form by a continuous screw-function coordinate. Let g be the screw function associated with zeta. On $(-a, a)$ one obtains a localized quadratic form $\mathcal{Q}_{W_a}$ and a self-adjoint operator A_a associated with it. The nonlocal first-order realization provides a concrete operator-theoretic interpretation of the finite intervals.

Theorem 3.1 (Localized self-adjoint Suzuki realization [9, 11]). *For each finite interval $(-a, a)$, the localized Weil quadratic form has a self-adjoint operator realization A_a . The construction is unconditional. Suzuki’s limiting conjecture states that the imaginary parts of the nontrivial zeros arise from the strong-resolvent limit of these finite-interval nonlocal realizations as $a \rightarrow \infty$.*

Remark 3.2. This theorem supplies the correct analytic language for convergence: strong resolvent convergence of finite-interval operators. It does not prove the limit that would put the full zero divisor on the line.

4. FINITE-TO-GLOBAL BRANCH CONVERGENCE

There are two different statements which must not be identified.

Definition 4.1 (Divisor convergence). Let ϕ_L be the actual selected continuum ground branch obtained from the finite radical quotients after $N \rightarrow \infty$ at fixed L . Divisor convergence means that, locally uniformly away from zeros,

$$(4.1) \quad \partial_z^2 \log \widehat{\phi}_L(z) \longrightarrow \partial_z^2 \log \Xi(z).$$

Definition 4.2 (Positive-form convergence). Positive-form convergence means that for every $v \in C_c^\infty(\mathbb{R})$ there are finite-level approximants $v_{L,N} \in V_{L,N}$ with

$$(4.2) \quad v_{L,N} \rightarrow v$$

in a topology strong enough for the explicit formula, and

$$(4.3) \quad \mathcal{Q}_W(v, v) \geq \limsup_{L \rightarrow \infty} \limsup_{N \rightarrow \infty} \left\langle T_{L,N} v_{L,N}, v_{L,N} \right\rangle - o(1).$$

The finite forms on the right side are nonnegative after shift and quotient. Thus positive-form convergence immediately implies Weil positivity. Divisor convergence implies the same conclusion if it is proved with enough control to exclude loss of spectral mass and branch switching.

Lemma 4.3 (Logarithmic-curvature branch form). *Let k_L be Connes' prolate model and ϕ_L the actual selected ground branch. Since $\widehat{k}_L \rightarrow \Xi$, divisor convergence is equivalent, on every simply connected compact set avoiding zeros, to*

$$(4.4) \quad \partial_z^2 \log \frac{\widehat{\phi}_L(z)}{\widehat{k}_L(z)} \longrightarrow 0.$$

Proof. Choose compatible logarithms on the compact set. Then

$$(4.5) \quad \partial_z^2 \log \widehat{\phi}_L - \partial_z^2 \log \Xi = \partial_z^2 \log \frac{\widehat{\phi}_L}{\widehat{k}_L} + \partial_z^2 \log \frac{\widehat{k}_L}{\Xi}.$$

The last term tends to zero by local uniform convergence and Cauchy estimates. This proves the equivalence. $\square$

5. CONDITIONAL CLOSURE OF THE CCM ROUTE

Theorem 5.1 (Connes closure theorem). *Assume the finite CCM package and Suzuki's localized self-adjoint realization. If either positive-form convergence or divisor convergence with no branch loss holds, then RH follows.*

Proof. Under positive-form convergence, the finite shifted quotient forms are nonnegative:

$$(5.1) \quad \langle T_{L,N} u, u \rangle \geq 0 \quad (u \perp \xi_{L,N}).$$

Given $v \in C_c^\infty(\mathbb{R})$, choose approximants $v_{L,N}$ as in (4.2)–(4.3). Passing to the limit gives

$$(5.2) \quad \mathcal{Q}_W(v, v) \geq 0.$$

By Theorem 1.1, RH follows.

Under divisor convergence with no branch loss, the limiting spectral divisor of the selected self-adjoint finite operators is the divisor of Ξ . Since every finite spectral parameter is real, the limiting zeros are of the form $1/2 + it$. Hence all zeros of ξ lie on the critical line. This is RH. $\square$

Remark 5.2. This theorem is the mathematical shape of the Connes proof. Everything finite is self-adjoint and critical-line. The only possible failure is at the infinite passage: positivity or spectral reality may be lost if the finite branch does not converge to the correct global branch with sign.

6. LOCALIZED REALIZATION AND THE GLOBAL LIMIT

Suzuki's screw-function operator provides the natural candidate topology for the limiting passage. In the notation of Theorem 3.1, the desired operator statement is

$$(6.1) \quad A_a \longrightarrow A_\infty$$

in strong resolvent sense, with

$$(6.2) \quad \text{Spec}(A_\infty) = \{\gamma : \xi(1/2 + i\gamma) = 0\}$$

including multiplicities.

Proposition 6.1 (Suzuki-to-Connes transfer). *If the CCM quotient operators and the Suzuki finite-interval operators are identified by a common core and their resolvents differ by $o(1)$ strongly on that core, then Suzuki's strong-resolvent limit implies the divisor convergence needed in Theorem 5.1.*

Proof. Strong resolvent convergence determines spectral projections at continuity sets of the limiting spectral measure. The common-core identification transfers those projections to the CCM branch. Since the finite CCM operators are self-adjoint, the limiting spectral measure is real. If its spectral determinant is the zeta-regularized determinant of Ξ , then the divisor of Ξ is supported on the real spectral line. $\square$

Remark 6.2. This proposition isolates a second version of the same branch-selection step: one can prove it as branch convergence of transforms, as form convergence, or as strong-resolvent convergence plus determinant identification.

7. BRANCH SELECTION AS A FINITE SOURCE THEOREM

The finite critical-line property and the prolate approximation do not alone prove RH. The branch-selection assertion is a signed convergence theorem. In the notation above it can be stated in three equivalent operational forms.

Problem 7.1 (Branch selector). Let ϕ_L be the actual ground branch selected by the finite CCM radical quotient after $N \rightarrow \infty$. Prove that

$$(7.1) \quad \partial_z^2 \log \frac{\widehat{\phi}_L(z)}{\widehat{k}_L(z)} \longrightarrow 0$$

locally uniformly on compact subsets of the open critical strip avoiding zeros.

Problem 7.2 (Positive finite-to-global transfer). For every $v \in C_c^\infty(\mathbb{R})$, construct finite quotient approximants $v_{L,N}$ for which the completed Weil form satisfies

$$(7.2) \quad \mathcal{Q}_W(v, v) = \lim_{L \rightarrow \infty} \lim_{N \rightarrow \infty} \langle T_{L,N} v_{L,N}, v_{L,N} \rangle + R(v),$$

with $R(v) \geq 0$, or at least with $R(v)$ controlled so that the right side is nonnegative.

Problem 7.3 (All-source sharp Doob floor). Let $K > 0$ be Riemann's full kernel, normalized by $\widehat{K} = \Xi$, and set

$$(7.3) \quad d\mu_K(x) = 2 \cosh(x/2) K(x) \, dx.$$

Let $\mathcal{E}_K$ be the complete jump energy containing all ordinary von Mangoldt atoms and the archimedean Gamma jump measure. Prove, directly from the literal sources, the sharp inequality

$$(7.4) \quad \boxed{\mathcal{E}_K(r) \geq \frac{1}{2} \text{Var}_{\mu_K}(r)}$$

for every even compactly supported smooth r .

Proposition 7.4 (Equivalence of the three gates). *Each of Problems 7.1, 7.2, and 7.3, together with the already established finite CCM–Suzuki components, implies RH. Conversely, RH implies each gate in the expected limiting sense.*

Proof. Problem 7.1 gives divisor convergence by Lemma 4.3, hence RH by Theorem 5.1. Problem 7.2 gives Weil positivity directly, hence RH by Theorem 1.1. Problem 7.3 is the exact ground-state form of Weil positivity:

$$(7.5) \quad \mathcal{Q}_W(Kr, Kr) = \mathcal{E}_K(r) - \frac{1}{2} \text{Var}_{\mu_K}(r).$$

Since every even compactly supported test has the form Kr , this proves the even Weil criterion, which is equivalent to RH. Under RH, Weil positivity holds; the finite critical-line branches then have no off-line divisor to converge to, and the same identities give the three gates in their limiting form. $\square$

8. FINITE BRANCH-SELECTION REFINEMENT

The gate which belongs to the Connes route is not the full-kernel inequality (7.4), because that has already eliminated the finite semilocal architecture. The required finite comparison is instead a one-sided Mosco transfer for the finite shifted forms, in the sense of Mosco convergence of closed forms [29, 30, 31]. We use Mosco only as a directional lower-transfer requirement; known instability phenomena for global path properties under Mosco convergence are a warning not to infer additional global structure from this step alone [32].

Fix $v \in C_c^\infty(\mathbb{R})$. Let $P_{L,N}v \in V_{L,N}$ be the Fourier sampling used in the CCM finite model, and remove the finite radical by

$$(8.1) \quad u_{L,N}(v) = P_{L,N}v - \frac{\langle P_{L,N}v, \xi_{L,N} \rangle}{\langle \xi_{L,N}, \xi_{L,N} \rangle} \xi_{L,N}.$$

Every object in (8.1) is finite and is defined before any reference to the zero locations of ζ .

Definition 8.1 (Finite transfer defect). The finite transfer defect of v is

$$(8.2) \quad \mathfrak{D}_{L,N}(v) = \mathcal{Q}_W(v, v) - \langle T_{L,N}u_{L,N}(v), u_{L,N}(v) \rangle.$$

Expanding both terms by the explicit formula and by the semilocal trace formula decomposes it as

$$(8.3) \quad \mathfrak{D}_{L,N} = \mathfrak{D}_{L,N}^{\text{mesh}} + \mathfrak{D}_{L,N}^{\text{ext}} + \mathfrak{D}_{L,N}^\Gamma + \mathfrak{D}_{L,N}^p + \mathfrak{D}_{L,N}^{\text{pole}} + \mathfrak{D}_{L,N}^{\text{br}}.$$

Here $\mathfrak{D}^{\text{mesh}}$ is Fourier discretization, $\mathfrak{D}^{\text{ext}}$ is the exterior lattice factor, $\mathfrak{D}^\Gamma$, $\mathfrak{D}^p$, and $\mathfrak{D}^{\text{pole}}$ are the literal Gamma, von Mangoldt and polar source differences, and $\mathfrak{D}^{\text{br}}$ is the finite radical branch correction.

Proposition 8.2 (Known analytic part of the transfer). *For each fixed $v \in C_c^\infty(\mathbb{R})$ there is a cofinal diagonal $N = N(L)$, with $N(L)/L^2 \rightarrow \infty$, such that*

$$(8.4) \quad \mathfrak{D}_{L,N(L)}^{\text{mesh}}(v) + \mathfrak{D}_{L,N(L)}^{\text{ext}}(v) \rightarrow 0.$$

The assertion uses only Fourier approximation, the exterior-lattice curvature bound, and the fixed- L cofinal summability already available in the CCM quotient calculation.

Proof. The Fourier sampling of a fixed compactly supported smooth test converges rapidly on every fixed physical interval. The exterior determinant curvature is $O_K(L^2/N)$ on compact spectral sets; the assumed diagonal therefore removes it. Fixed- L cofinal summability supplies the remaining quotient-to-continuum passage before $L \rightarrow \infty$. $\square$

Problem 8.3 (Source-side Mosco lower bound). Prove that for every $v \in C_c^\infty(\mathbb{R})$ one can choose the cofinal diagonal of Proposition 8.2 so that

$$(8.5) \quad \liminf_{L \rightarrow \infty} \left(\mathfrak{D}_{L,N(L)}^\Gamma(v) + \mathfrak{D}_{L,N(L)}^p(v) + \mathfrak{D}_{L,N(L)}^{\text{pole}}(v) + \mathfrak{D}_{L,N(L)}^{\text{br}}(v) \right) \geq 0.$$

Theorem 8.4 (Mosco lower transfer closes the Connes route). *If Problem 8.3 holds, then the positive finite CCM forms transfer to the completed Weil form. Consequently RH follows.*

Proof. For the diagonal in Proposition 8.2 and Problem 8.3, equations (8.2)–(8.5) give

$$(8.6) \quad \mathfrak{D}_{L,N(L)}(v) \geq -o(1).$$

Since $T_{L,N} \geq 0$ on the finite quotient, (8.2) implies

$$(8.7) \quad \mathcal{Q}_W(v, v) = \langle T_{L,N(L)}u_{L,N(L)}(v), u_{L,N(L)}(v) \rangle + \mathfrak{D}_{L,N(L)}(v) \geq -o(1).$$

Letting $L \rightarrow \infty$ gives $\mathcal{Q}_W(v, v) \geq 0$. Since v was arbitrary, Weil positivity follows. Theorem 1.1 then gives RH. $\square$

Remark 8.5. Problem 8.3 is the next target in the Connes path. It is not the classical full-kernel inequality (7.4). It is a finite-source statement: the Gamma term, the literal von Mangoldt atoms, the pole and the radical branch correction must be kept together until after the lower limit is taken. Separating them reproduces the previously encountered sign obstructions.

Remark 8.6 (Joint-source rule). The source-side lower bound is pursued below through the complete finite prime–Gamma form and its radical anti-short. The archimedean, polar, ordinary-prime and radical contributions may exchange mass, so no separate lower bound is imposed on any one of them. The active finite statement is the joint inertia/gain inequality developed in Section 9.

8.1. Controlled exterior and cross residuals. We now specify the moving vector used in the closure package and discharge the even Rayleigh residual. This specification is essential: a trial vector satisfying only the two co-Poisson moment conditions may retain a nonzero endpoint value and hence a literal t^{-1} exterior carrier. The following fixed six-mode construction also removes the first two exterior boundary symbols without changing the d_4, d_8 angle ladder.

Let

$$J_6 = \{0, 4, 8, 12, 16, 20\}$$

and let $\psi_{j,\lambda}$ be the real normalized even prolate spheroidal wave functions in the Fourier +1 sector. Write

$$(8.8a) \quad a_j = \psi_{j,\lambda}(0), \quad e_j = \psi_{j,\lambda}(\lambda), \quad q_j = 1 - \chi_j, \quad d_j = 1 - \chi_j^2,$$

where χ_j is the corresponding finite Fourier eigenvalue, and let μ_j be the separation constant in the prolate differential equation.

Definition 8.7 (Endpoint-corrected moving source). Set

$$(8.8b) \quad \mathcal{K}_\lambda^{\text{ep}} = \left\{ f = \sum_{j \in J_6} c_j \psi_{j,\lambda} : f(0) = 0, \quad \int_{\mathbb{R}} f = 0, \quad f(\lambda) = f'(\lambda) = 0 \right\}.$$

Let f_λ^{ep} be a normalized first generalized eigenvector of $\sum d_j |c_j|^2$ on this space. Apply the co-Poisson construction and positive-parity projection of Section 8 to f_λ^{ep} , and denote the resulting interior and exterior vectors by V_λ^+ and R_λ^+ . Henceforth the moving vector is

$$(8.8c) \quad q_L^+ = \frac{V_\lambda^+}{\|V_\lambda^+\|}.$$

This is also the vector used in the cross residual of Theorem 8.12.

Lemma 8.8 (The endpoint constraints preserve the angle ladder). *For all sufficiently large λ , the four constraints in (8.8b) are independent. If $\delta_{0,\lambda}^{\text{ep}} \leq \delta_{1,\lambda}^{\text{ep}}$ are the generalized eigenvalues of $\sum d_j |c_j|^2$ relative to $\sum |c_j|^2$ on $\mathcal{K}_\lambda^{\text{ep}}$, then*

$$(8.8d) \quad \delta_{0,\lambda}^{\text{ep}} \asymp d_4, \quad \delta_{1,\lambda}^{\text{ep}} \asymp d_8.$$

Moreover

$$(8.8e) \quad \|V_\lambda^+\| \geq c_V > 0$$

with c_V independent of λ .

Proof. The exact endpoint-slope identity for a prolate linear combination is

$$(8.8f) \quad 2\lambda f'(\lambda) = \sum_{j \in J_6} c_j (\mu_j - 4\pi^2 \lambda^4) e_j.$$

After imposing $f(\lambda) = 0$, the four constraint rows are therefore

$$(8.8g) \quad (a_j), \quad (q_j a_j), \quad (e_j), \quad (\mu_j e_j).$$

For $j = 4r$, $0 \leq r \leq 5$, the fixed-order prolate-to-Hermite and endpoint asymptotics give

$$(8.8h) \quad |a_{4r}| \asymp 1, \quad |e_{4r}| \asymp \lambda^{1/2} q_{4r}^{1/2}, \quad \mu_{4r} = \lambda^2 (M_0 + M_1 r + O(\lambda^{-2})),$$

where $M_1 \neq 0$, together with

$$(8.8i) \quad q_{4(r+1)}/q_{4r} \asymp \lambda^8.$$

After multiplication by uniformly invertible diagonal row and column scalings, and with $y = \lambda^4$, the constraint matrix is a uniform perturbation of

$$(8.8j) \quad \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & y^2 & y^4 & y^6 & y^8 & y^{10} \\ 1 & y & y^2 & y^3 & y^4 & y^5 \\ 0 & y & 2y^2 & 3y^3 & 4y^4 & 5y^5 \end{pmatrix}.$$

Its consecutive four-column minors are nonzero. Elimination gives two kernel profiles, up to uniformly bounded nonzero factors,

$$(8.8k) \quad \begin{aligned} v^{(0)} &= (1, -1, 2y^{-1}, -y^{-2}, y^{-4}, 0) + O(y^{-2}), \\ v^{(1)} &= (0, 1, -1, 2y^{-1}, -y^{-2}, y^{-4}) + O(y^{-2}). \end{aligned}$$

Since $d_{4r} \asymp d_4 y^{2(r-1)}$, the quotient of the first profile is $\asymp d_4$. After removing that direction, the quotient of the second is $\asymp d_8$. The pivot equations and Cauchy–Schwarz give the matching lower bounds; the perturbation in (8.8j) is absorbed by the min–max principle. This proves (8.8d). Finally, the first profile has the same nonzero 0/4 Hermite limit as the original moving vector. Positive-parity co-Poisson projection is continuous on this fixed-mode limit, which proves (8.8e). $\square$

Put

$$(8.8l) \quad \ell_\lambda = (1 - P_\lambda) \hat{f}_\lambda^{\text{ep}}.$$

Lemma 8.9 (Uniform exterior two-jet estimate). *Uniformly for $t \geq \lambda$,*

$$(8.8m) \quad |\ell_\lambda(t)| + \lambda^{-1} |\ell'_\lambda(t)| \ll \lambda^{7/2} \sqrt{d_4} t^{-3}.$$

Proof. Put $c = 2\pi\lambda^2$ and $x = t/\lambda$. The Fourier continuation of each fixed prolate mode is its radial PSWF. We apply the uniform Bessel and Liouville–Green expansions of Dunster [19], including their differentiated error bounds.

For $1 \leq x \leq 2$, the uniform Bessel expansion and Lemma 8.8 give

$$(8.8n) \quad |\ell_\lambda(\lambda x)| + \lambda^{-1} |\ell'_\lambda(\lambda x)| \ll \lambda^{1/2} \sqrt{d_4},$$

which implies (8.8m) because $x^{-3} \geq 1/8$.

For $2 \leq x \leq c$, specialize Dunster’s expansion to fixed order $j \in J_6$. Its small parameter satisfies

$$(8.8o) \quad \sigma_j^2 = \frac{2j+1}{c} + O(c^{-2}).$$

All indices are congruent modulo four, so the term $-j\pi/2$ has the same outgoing orientation. The Liouville phase has the expansion

$$(8.8p) \quad \xi(\sigma, x) + J(\sigma) = \sqrt{x^2 - 1} + \sigma^2 \left(\frac{\pi}{4} - \frac{1}{2} \operatorname{arcsec} x \right) + O(\sigma^4(1 + x^{-1})).$$

Consequently the radial expansion through order c^{-1} can be written

$$(8.8q) \quad \widehat{\psi}_{j,\lambda}(\lambda x) = e_j(U_{0,\lambda}(x) + \mu_j U_{1,\lambda}(x)) + E_{j,\lambda}(x),$$

where the two U 's are independent of j , and the uniform Olver remainder, also after one derivative, is

$$(8.8r) \quad |E_{j,\lambda}(x)| + \lambda^{-1} |\partial_t E_{j,\lambda}(x)| \ll |e_j| (x^{-3} + c^{-2} x^{-1}).$$

Indeed, (8.8p) makes the mode-dependent phase $(2j+1)/(2x) + O_j(x^{-3} + c^{-1}x^{-1})$, while the radial amplitude is $x^{-1}(1 + O_j(x^{-2} + c^{-1}))$. Thus its constant and affine terms in $2j+1$ are precisely the two symbols in (8.8q). They cancel because

$$(8.8s) \quad \sum_j c_j e_j = 0, \quad \sum_j c_j \mu_j e_j = 0$$

by the endpoint constraints. Since $\sum_j |c_j e_j| \ll \lambda^{1/2} \sqrt{d_4}$ and $c^{-2} x^{-1} \leq x^{-3}$ for $x \leq c$, summing (8.8q)–(8.8r) proves (8.8m) throughout this region.

For completeness, the same cancellation controls the range $x \geq c$ without extrapolating the middle expansion. If the outgoing solution is written

$$y_j(t) = e^{2\pi i \lambda t} \sum_{m \geq 0} a_{m,j} t^{-m-1} + \text{conjugate},$$

then, with $A_j = 4\pi^2 \lambda^4 - \mu_j$, its exact recurrence is

$$(8.8t) \quad \begin{aligned} 2i(2\pi\lambda) m a_{m,j} &= [m(m-1) + A_j] a_{m-1,j} \\ &+ 2i(2\pi\lambda) \lambda^2 (m-1) a_{m-2,j} \\ &- \lambda^2 (m-2)(m-1) a_{m-3,j}. \end{aligned}$$

The first two combined coefficients vanish by (8.8s), and

$$(8.8u) \quad \sum_j c_j a_{2,j} = -\frac{1}{8(2\pi\lambda)^2} \sum_j c_j \mu_j^2 a_{0,j}.$$

The resulting Volterra majorant, using $|\mu_j| \ll \lambda^2$, gives

$$(8.8v) \quad |\ell_\lambda(t)| + \lambda^{-1} |\ell'_\lambda(t)| \ll \lambda^{5/2} \sqrt{d_4} t^{-3}, \quad x \geq c,$$

which is stronger than (8.8m). The three regions cover $[\lambda, \infty)$. $\square$

Theorem 8.10 (Even Rayleigh residual). *For the endpoint-corrected moving vector of Definition 8.7,*

$$(8.8w) \quad \boxed{|\mathcal{R}_L^+| \ll \lambda^7 d_4.}$$

Thus the first residual exponent is $p_R = 7 < 8$.

Proof. Define, for $v \geq \lambda$,

$$(8.8x) \quad H_\lambda(v) = v^{1/2} \sum_{n \geq 1} \ell_\lambda(nv), \quad D = v \partial_v.$$

Lemma 8.9 and absolute summation imply

$$(8.8y) \quad |H_\lambda(v)| \ll \lambda^{7/2} \sqrt{d_4} v^{-5/2}, \quad |DH_\lambda(v)| \ll \lambda^{9/2} \sqrt{d_4} v^{-3/2}.$$

For $|\Re w| < 1/2$, put

$$(8.8z) \quad A_\lambda(w) = \int_\lambda^\infty H_\lambda(v) v^{-w} \frac{dv}{v}.$$

The direct estimate for bounded ordinate and the identity

$$(8.8aa) \quad wA_\lambda(w) = H_\lambda(\lambda) \lambda^{-w} + \int_\lambda^\infty DH_\lambda(v) v^{-w} \frac{dv}{v}$$

for large ordinate yield, uniformly in the strip,

$$(8.8ab) \quad |A_\lambda(w)| \ll \frac{\lambda^{7/2} \sqrt{d_4}}{1 + |\Im w|}.$$

The exact positive-parity Mellin coordinate is

$$(8.8ac) \quad b_\lambda(w) = \frac{A_\lambda(w) + A_\lambda(-w)}{\sqrt{2}},$$

and the zero-side explicit formula gives

$$(8.8ad) \quad QW(R_\lambda^+, R_\lambda^+) = \sum_\rho b_\lambda(\rho - \tfrac{1}{2})^2.$$

The local zero-count estimate $N(T+1) - N(T) \ll \log(2+T)$, together with (8.8ab), makes this series absolutely convergent and gives

$$(8.8ae) \quad |QW(R_\lambda^+, R_\lambda^+)| \ll \lambda^7 d_4.$$

Finally, the exact normalized residual identity and (8.8e) give

$$|\mathcal{R}_L^+| = \frac{|QW(R_\lambda^+, R_\lambda^+)|}{\|V_\lambda^+\|^2} \ll \lambda^7 d_4,$$

as claimed. □

Lemma 8.11 (Dyadic complex-frequency Bessel bound). *Let $I_L = [-L/2, L/2]$, $\lambda = e^{L/2}$, and let coefficients indexed by the nontrivial zeta divisor satisfy*

$$(8.8af) \quad |a_\rho| \leq \frac{B}{1 + |\gamma|}, \quad \rho = \beta + i\gamma.$$

Then the symmetric partial sums of

$$(8.8ag) \quad h(x) = \sum_\rho a_\rho e^{-i\gamma x} e^{(\beta-1/2)x}$$

converge in $L^2(I_L)$, and

$$(8.8ah) \quad \boxed{\|h\|_{L^2(I_L)} \ll B \sqrt{\lambda(1+L)}}.$$

The implied constant is absolute for the Riemann divisor.

Proof. It is enough to treat the positive ordinates, since the negative ordinates are obtained by conjugation. Separate a bounded initial block and put

$$(8.8ai) \quad \mathcal{Z}_k = \{\rho : 2^k \leq \gamma < 2^{k+1}\}, \quad k \geq 0.$$

For $\rho, \rho' \in \mathcal{Z}_k$, direct integration gives the Gram entry

$$(8.8aj) \quad G_{\rho, \rho'} = \int_{-L/2}^{L/2} e^{(\beta+\beta'-1)x} e^{-i(\gamma-\gamma')x} dx,$$

$$|G_{\rho, \rho'}| \ll \lambda \min\left(L, \frac{1}{|\gamma - \gamma'|}\right),$$

where the second expression is read as L at coincident ordinates. Partition the ordinate axis into unit intervals. The unconditional local zero count

$$(8.8ak) \quad N(T+1) - N(T) \ll \log(3+T)$$

shows that the interval containing γ , together with its two neighbors, contributes $O(\lambda L(1+k))$ to a Gram row. The unit intervals at distance $m \geq 2$ contribute $O(\lambda(1+k)/m)$. Hence

$$(8.8al) \quad \sup_{\rho \in \mathcal{Z}_k} \sum_{\rho' \in \mathcal{Z}_k} |G_{\rho, \rho'}| \ll \lambda(1+L+k)(1+k).$$

Schur's test and Riemann–von Mangoldt now give

$$(8.8am) \quad \left\| \sum_{\rho \in \mathcal{Z}_k} a_\rho e^{-i\gamma x} e^{(\beta-1/2)x} \right\|_2^2 \ll \lambda(1+L+k)(1+k) \sum_{\rho \in \mathcal{Z}_k} |a_\rho|^2,$$

$$\sum_{\rho \in \mathcal{Z}_k} |a_\rho|^2 \ll B^2 \frac{1+k}{2^k}.$$

After taking square roots, the triangle inequality over the dyadic blocks yields

$$(8.8an) \quad \|h\|_2 \ll B\sqrt{\lambda} \sum_{k \geq 0} \frac{(1+k)\sqrt{1+L+k}}{2^{k/2}} \ll B\sqrt{\lambda(1+L)}.$$

The same summable majorant makes the symmetric partial sums Cauchy in $L^2(I_L)$, proving the lemma. $\square$

Theorem 8.12 (Cross residual for the direct branch selector). *For the endpoint-corrected moving vector of Definition 8.7,*

$$(8.8ao) \quad \boxed{\|b_L^+\| \ll \lambda^4 \sqrt{(1+L)d_4}.}$$

In particular, $\|b_L^+\| \rightarrow 0$.

Proof. Put

$$(8.8ap) \quad e_L^+ = \frac{R_\lambda^+}{\|V_\lambda^+\|}.$$

The exact moving-radical identity gives, for every even core vector $g \perp q_L^+$,

$$(8.8aq) \quad \langle b_L^+, g \rangle = -\mathcal{Q}_W(e_L^+, g).$$

If $z_\rho = \rho - 1/2$, the positive-parity Mellin coordinate is

$$(8.8ar) \quad \widehat{R}_\lambda^+(z_\rho) = b_\lambda(z_\rho), \quad b_\lambda(w) = \frac{A_\lambda(w) + A_\lambda(-w)}{\sqrt{2}}.$$

Equations (8.8ab), (8.8e), and $0 < \Re \rho < 1$ therefore give the complete-divisor representation

$$(8.8as) \quad \mathcal{Q}_W(e_L^+, g) = \sum_{\rho} c_{\rho, \lambda} \widehat{g}(z_\rho), \quad |c_{\rho, \lambda}| \ll \frac{B_\lambda}{1+|\gamma|}, \quad B_\lambda = \lambda^{7/2} \sqrt{d_4}.$$

Lemma 8.11 shows that this functional has an $L^2(I_L)$ Riesz representative h_λ satisfying

$$(8.8at) \quad \|h_\lambda\|_2 \ll B_\lambda \sqrt{\lambda(1+L)} = \lambda^4 \sqrt{(1+L)d_4}.$$

Taking the supremum of (8.8aq) over the even unit ball orthogonal to q_L^+ proves (8.8ao). Finally,

$$(8.8au) \quad d_4 \asymp \lambda^9 e^{-4\pi\lambda^2}$$

implies

$$(8.8av) \quad \lambda^4 \sqrt{(1+L)d_4} \asymp \lambda^{17/2} \sqrt{1+L} e^{-2\pi\lambda^2} \longrightarrow 0.$$

□

Theorem 8.12 closes the cross residual required by the direct asymptotic-inertia criterion. Together with Theorem 8.10, it gives

$$(8.8aw) \quad |\mathcal{R}_L^+| + \|b_L^+\| \longrightarrow 0.$$

9. COFINAL PRIME–GAMMA INERTIA MATCHING

The preceding moving-source analysis proves the two residual estimates. The individual source components can exchange mass. We therefore keep all sources in one finite Hermitian form.

9.1. Complete and finite joint heads. On the centered real-even space

$$(9.1) \quad \mathcal{H}_0 = \left\{ q \in L^2(\mu_K) : \int q \, d\mu_K = 0 \right\},$$

define the polarized displacement atom

$$(9.2) \quad \begin{aligned} \mathcal{J}_u(q, s) &= \int_{\mathbb{R}} K(x) K(x-u) \overline{q(x) - q(x-u)} \\ &\quad \times (s(x) - s(x-u)) \, dx. \end{aligned}$$

For $X \geq 2$, put

$$(9.3) \quad \begin{aligned} \mathcal{A}_X(q, s) &= \mathcal{E}_\Gamma(q, s) + \sum_{p^k \leq X} \frac{\Lambda(p^k)}{p^{k/2}} \mathcal{J}_{k \log p}(q, s) - \frac{1}{2} \langle q, s \rangle_{\mu_K}, \\ \mathcal{T}_X(q, s) &= \sum_{p^k > X} \frac{\Lambda(p^k)}{p^{k/2}} \mathcal{J}_{k \log p}(q, s). \end{aligned}$$

The complete form is

$$(9.4) \quad \mathcal{A}_\infty = \mathcal{E}_K - \frac{1}{2} \langle \cdot, \cdot \rangle_{\mu_K}, \quad \boxed{\mathcal{A}_X = \mathcal{A}_\infty - \mathcal{T}_X, \quad \mathcal{T}_X \succeq 0.}$$

Thus the finite heads increase in Loewner order toward the complete form.

Let $V_M \subset \mathcal{H}_0$ be a nested finite-dimensional mode space in the complement of the complete radical. The elementary choice is generated by centered projections of

$$(9.5) \quad \chi_z(x) = \frac{\cos(zx)}{\cosh(x/2)}$$

for complete zero orbits z in a growing spectral rectangle, with all multiplicity jets retained. Let $Z_M : \mathbb{C}^{d_M} \rightarrow V_M$ be the synthesis map, and write

$$(9.6) \quad \mathbf{N}_M = Z_M^* Z_M, \quad \mathbf{H}_{M,X} = Z_M^* \mathcal{A}_X Z_M.$$

Theorem 9.1 (Cofinal filter-bank approximation). *There are constants $c > 0$ and $C_M < \infty$ such that, for $Y \geq X$,*

$$(9.7) \quad 0 \preceq \mathbf{N}_M^{-1/2}(\mathbf{H}_{M,\infty} - \mathbf{H}_{M,Y})\mathbf{N}_M^{-1/2} \preceq \mathbf{N}_M^{-1/2}(\mathbf{H}_{M,\infty} - \mathbf{H}_{M,X})\mathbf{N}_M^{-1/2} \preceq C_M e^{-cX} I.$$

Consequently, for arbitrary $\varepsilon_M \downarrow 0$, the choice

$$(9.8) \quad X(M) \geq \frac{1}{c} \left(\log C_M + \log \frac{1}{\varepsilon_M} \right)$$

makes the omitted literal Euler bank at most ε_M in normalized operator norm.

Proof. The first two inequalities follow from (9.3): each omitted prime-power atom is a positive Gram form. For fixed M , every mode in (9.5) has imaginary part strictly inside the critical strip. The theta expansion of $K(x)K(x-u)$, with $u = \log p^k$, is therefore bounded on V_M by a fixed polynomial times $\exp(-cp^k)$. Summation over prime powers larger than X , followed by conjugation with $\mathbf{N}_M^{-1/2}$, gives the last inequality. Solving $C_M e^{-cX} \leq \varepsilon_M$ gives (9.8). $\square$

Define the normalized bank $S_{M,X}$ by

$$(9.9) \quad S_{M,X}^* S_{M,X} = \mathbf{N}_M^{-1/2} \left(\mathcal{E}_\Gamma + \sum_{p^k \leq X} \frac{\Lambda(p^k)}{p^{k/2}} \mathcal{J}_{k \log p} \right)_{V_M} \mathbf{N}_M^{-1/2}.$$

Then the exact signed filter-bank identity is

$$(9.10) \quad \mathbf{G}_{M,X} := \mathbf{N}_M^{-1/2} \mathbf{H}_{M,X} \mathbf{N}_M^{-1/2} = S_{M,X}^* S_{M,X} - \frac{1}{2} I.$$

No scalar averaging occurs in (9.10): all modes and all retained sources remain coupled in one matrix.

9.2. The strict finite frontier. Put

$$(9.11) \quad \delta_M = \lambda_{\min} \left(\mathbf{N}_M^{-1/2} \mathbf{H}_{M,\infty} \mathbf{N}_M^{-1/2} \right), \quad X_*(M) = \inf \{ X : \mathbf{H}_{M,X} \succeq 0 \}.$$

Theorem 9.2 (Exact finite-repair criterion). *For every fixed M ,*

$$(9.12) \quad X_*(M) < \infty \iff \delta_M > 0.$$

If $\delta_M > 0$, then

$$(9.13) \quad X_*(M) \leq \left\lceil \frac{1}{c} \log \frac{C_M}{\delta_M} \right\rceil.$$

If $\delta_M = 0$, every proper finite head has a negative direction and its least eigenvalue increases to 0 without crossing. If $\delta_M < 0$, no finite head repairs the completed negative direction.

Proof. If $\delta_M > 0$, Theorem 9.1 gives

$$\mathbf{H}_{M,X} \succeq (\delta_M - C_M e^{-cX}) \mathbf{N}_M,$$

which proves (9.13). Conversely, suppose $\mathbf{H}_{M,X} \succeq 0$. The omitted tail is positive definite on V_M : if it vanished on a nonzero q , then the displacement atoms of two omitted distinct primes would make q periodic with periods $\log p, \log r$. Their ratio is irrational, so continuity makes q constant; centeredness then makes $q = 0$, a contradiction. Hence

$$\mathbf{H}_{M,\infty} = \mathbf{H}_{M,X} + (\mathbf{H}_{M,\infty} - \mathbf{H}_{M,X}) \succ 0.$$

Finite dimensionality gives $\delta_M > 0$. Evaluating a finite head on a minimizer of (9.11) proves the last two assertions. $\square$

Remark 9.3 (What discreteness proves). The finite spectra in (9.11) are discrete, but discreteness alone does not imply (9.12)'s left side. The scalar sequence $-e^{-cX}$ is already a discrete level which increases to 0 and never crosses it. The arithmetic theorem must therefore prove a strict completed gap or use the vanishing compensated frontier below.

9.3. Radical shorting and the inertia staircase. The word shorting is used in the Krein–Anderson–Trapp sense: one eliminates an indefinite or constrained block through the extremal shorted/compressed operator before reading the remaining Schur complement [20, 21, 22]. The radical itself is part of the Sonin/co-Poisson and Weil-distribution Hilbert-space background cited in Remark 1.2; the explicit finite coordinate $r_j = K^{(2j)}/K - 4^{-j}$ is the coordinate used here for the completed radical.

Use a distinct symbol for the exact completed radical:

$$(9.14) \quad r_j = \frac{K^{(2j)}}{K} - 4^{-j}, \quad \mathfrak{R}_J = \text{span}\{r_1, \dots, r_J\}.$$

Polarization of the radical identity gives

$$(9.15) \quad \mathcal{A}_\infty(r, f) = 0 \quad (r \in \overline{\cup_J \mathfrak{R}_J}).$$

In bases of V_M and $\mathfrak{R}_J$, write the omitted-tail blocks as $\mathbf{T}^{VV}, \mathbf{T}^{VR}, \mathbf{T}^{RR}$. The radical block is strictly positive:

$$(9.16) \quad \mathbf{T}^{RR} \succ 0.$$

Indeed, vanishing would again give two incommensurable prime periods, whereas the largest derivative in a nonzero finite combination of the r_j has nonconstant theta asymptotics.

The finite-head matrix is therefore

$$(9.17) \quad \mathbf{A}_{M,J,X} = \begin{pmatrix} \mathbf{H}_M - \mathbf{T}^{VV} & -\mathbf{T}^{VR} \\ -\mathbf{T}^{RV} & -\mathbf{T}^{RR} \end{pmatrix}, \quad \mathbf{H}_M = \mathbf{H}_{M,\infty},$$

and its Schur complement is

$$(9.18) \quad \mathbf{S}_{M,J,X} = \mathbf{H}_M - (\mathbf{T}^{VV} - \mathbf{T}^{VR}(\mathbf{T}^{RR})^{-1}\mathbf{T}^{RV}).$$

Theorem 9.4 (Finite inertia matching [23, 24]). *One has*

$$(9.19) \quad \boxed{n_-(\mathbf{A}_{M,J,X}) = J + n_-(\mathbf{S}_{M,J,X}).}$$

Consequently,

$$(9.20) \quad \boxed{n_-(\mathbf{A}_{M,J,X}) = J \iff \mathbf{T}^{VV} - \mathbf{T}^{VR}(\mathbf{T}^{RR})^{-1}\mathbf{T}^{RV} \preceq \mathbf{H}_M.}$$

Proof. Block Gaussian elimination is a congruence from (9.17) to $\mathbf{S}_{M,J,X} \oplus (-\mathbf{T}^{RR})$. Equations (9.16), Sylvester's law of inertia, and (9.18) give (9.19)–(9.20). $\square$

Define the excess negative index and its frontier by

$$(9.21) \quad \kappa(M, J, X) = n_-(\mathbf{A}_{M,J,X}) - J, \quad X_*(M, J) = \inf\{X : \kappa(M, J, X) = 0\}.$$

Loewner monotonicity and interlacing give [25, 26]

parameter increased	effect on κ
X	nonincreasing
M	nondecreasing
J	nonincreasing.

Thus $X_*(M+1, J) \geq X_*(M, J)$, whereas $X_*(M, J+1) \leq X_*(M, J)$. This is the exact mode–prime–radical staircase: a head ending at 7 can close one row and fail when the next mode is admitted.

9.4. Geometry of a large-prime channel. The staircase has a literal signal-processing interpretation. For χ_z in (9.5), uniformly on fixed compact z, w -blocks,

$$(9.23) \quad \boxed{\frac{\log p}{\sqrt{p}} \mathcal{J}_{\log p}(\chi_z, \chi_w) = \beta_p \overline{A_p(z)} A_p(w) + E_p(z, w),}$$

where

$$(9.24) \quad \begin{aligned} A_p(z) &= 2z \sin\left(\frac{z \log p}{2}\right) + \tanh\left(\frac{\log p}{4}\right) \cos\left(\frac{z \log p}{2}\right), \\ \beta_p &= C_{\Xi}^2 \pi^3 (\log p) p^2 e^{-2\pi p} (1 + p^{-1/2})^{-2}, \end{aligned}$$

and

$$(9.25) \quad |E_p(z, w)| \leq C_B \beta_p p^{b-1}, \quad b = \max(|\operatorname{Im} z|, |\operatorname{Im} w|) < \frac{1}{2}.$$

The proof is Laplace’s method at the midpoint $x = \frac{1}{2} \log p$: $K(x)K(x - \log p)$ has aperture $p^{-1/2}$, and the first odd Taylor coefficient of χ_z is exactly $A_p(z)$. Hence one large prime is an approximate rank-one derivative sensor. In its asymptotic regime, d newly exposed independent directions require at least d independent prime channels. This proves the moving-bank geometry but not, by itself, the uniform lower frame bound in (9.20).

9.5. Finite observability and the exact gain budget. The distinction between observability and gain can be made exact. Every single literal displacement observes every finite elementary mode space.

Theorem 9.5 (One-channel finite observability). *For every $u > 0$ and every nonzero $q \in V_M$,*

$$(9.25a) \quad \boxed{\mathcal{J}_u(q, q) > 0.}$$

Consequently the atom Gram matrix is positive definite on V_M . Its normalized least eigenvalue is bounded away from zero uniformly when u ranges over a fixed compact subinterval of $(0, \infty)$.

Proof. The real theta kernel is strictly positive. Therefore $\mathcal{J}_u(q, q) = 0$ would imply $q(x) = q(x - u)$ for almost every x , and analyticity extends the equality to the real line. Every finite linear combination of the modes (9.5), including confluent multiplicity jets, tends to zero at both ends because $|\operatorname{Im} z| < 1/2$. A periodic function with this decay is zero. This proves strict positivity. Continuity in u and compactness of the normalized unit sphere give the uniform assertion. $\square$

Write the normalized Gamma deficit and prime bank as

$$(9.25b) \quad \mathbf{B}_M = \frac{1}{2}I - \mathbf{N}_M^{-1/2}(\mathcal{E}_\Gamma)_{V_M} \mathbf{N}_M^{-1/2}, \quad \mathbf{P}_{M,X} = \sum_{p^k \leq X} \frac{\log p}{p^{k/2}} \mathbf{N}_M^{-1/2}(\mathcal{J}_{k \log p})_{V_M} \mathbf{N}_M^{-1/2}.$$

Then $\mathbf{G}_{M,X} = \mathbf{P}_{M,X} - \mathbf{B}_M$, and congruence by $\mathbf{P}_{M,X}^{-1/2}$ gives

$$(9.25c) \quad \mathbf{G}_{M,X} \succeq 0 \iff \vartheta_{M,X} := \lambda_{\max}(\mathbf{P}_{M,X}^{-1/2} \mathbf{B}_M \mathbf{P}_{M,X}^{-1/2}) \leq 1.$$

Whenever a new prime-power atom is added, Theorem 9.5 makes $\vartheta_{M,X}$ strictly decrease on the positive-deficit sector. Norm convergence gives

$$(9.25d) \quad X_*(M) < \infty \iff \vartheta_{M,\infty} < 1.$$

Thus aliasing, finite rank and lack of sampling are no longer possible obstructions. The strict inequality in (9.25d) is the gain question; it cannot be replaced by the sum of the individual least atom eigenvalues, because that replacement discards the cross-prime alignment.

9.6. Scalar innovations of the completed staircase. There is an exact one-scalar recursion for the remaining matrix sign. Choose a basis $\phi_1, \phi_2, \dots$ adapted to the nested spaces and write

$$(9.25e) \quad \mathbf{H}_M = \begin{pmatrix} \mathbf{H}_{M-1} & c_M \\ c_M^* & h_M \end{pmatrix}.$$

Assuming $\mathbf{H}_{M-1} \succ 0$, set

$$(9.25f) \quad a_M = \mathbf{H}_{M-1}^{-1} c_M, \quad q_M^* = \phi_M - \sum_{j < M} (a_M)_j \phi_j.$$

Theorem 9.6 (Completed scalar innovation). *The vector q_M^* is orthogonal to V_{M-1} in the complete signed form and*

$$(9.25g) \quad \begin{aligned} \sigma_M &:= \mathcal{A}_\infty(q_M^*, q_M^*) = h_M - c_M^* \mathbf{H}_{M-1}^{-1} c_M = \frac{\det \mathbf{H}_M}{\det \mathbf{H}_{M-1}}, \\ \mathbf{H}_M \succ 0 &\iff \mathbf{H}_{M-1} \succ 0 \text{ and } \sigma_M > 0. \end{aligned}$$

Moreover the scalar retains the complete physical source decomposition

$$(9.25h) \quad \sigma_M = \mathcal{E}_\Gamma(q_M^*) + \sum_{p^k} \frac{\log p}{p^{k/2}} \mathcal{J}_{k \log p}(q_M^*, q_M^*) - \frac{1}{2} \|q_M^*\|_{\mu_K}^2.$$

Proof. Block multiplication gives $\mathcal{A}_\infty(v, q_M^*) = 0$ for $v \in V_{M-1}$ and the first line of (9.25g). Congruence by triangular block elimination reduces $\mathbf{H}_M$ to $\mathbf{H}_{M-1} \oplus [\sigma_M]$; Sylvester's law and the block determinant formula give the second line. Substitution in (9.3) gives (9.25h). $\square$

Before radical anti-shortening, the strict frontier is finite in every row precisely when all literal innovations (9.25h) are positive. This scalar criterion is an exact unshorted precursor: it keeps Gamma, the threshold, all ordinary von Mangoldt weights and their directional alignment in one number for each newly admitted mode. The active shortened target is identified below by $\mathfrak{g}_{M,J,X} > \delta_{M,J,X}$.

The moving-filter interpretation also has an exact update law. Suppose the preceding-mode block A is positive before the atom $n = p^k$ is inserted, write $w_n = \Lambda(n)/\sqrt{n}$, let $U_n = D_n|_{V_{M-1}}$, and put

$$(9.25i) \quad D_n q(x) = \sqrt{K(x)K(x - \log n)} \{q(x) - q(x - \log n)\}.$$

If $q_{M,X}^*$ is the adaptive finite-head regression residual and $r_n = D_n q_{M,X}^*$, direct Schur minimization gives

$$(9.25j) \quad \sigma_{M,+} - \sigma_{M,-} = w_n \langle r_n, (I + w_n U_n A^{-1} U_n^*)^{-1} r_n \rangle > 0.$$

Indeed, after writing an old-mode correction as y , the new pivot is the minimum of

$$\sigma_{M,-} + \langle y, Ay \rangle + w_n \|r_n + U_n y\|^2;$$

the normal equation and the Woodbury identity give (9.25j). Iterating over the ordered prime powers yields

$$(9.25k) \quad \sigma_{M,X_L} = \sigma_{M,X_0} + \sum_{X_0 < n_\ell \leq X_L} \Delta_{M,\ell}, \quad \Delta_{M,\ell} > 0.$$

Thus the row closes at the first index at which this exact cumulative innovation exceeds the initial deficit. The series converges to σ_M ; it cannot be replaced by a divergent “infinitely many channels” argument, because theta localization makes the large-prime gain summable.

Several primes can be kept jointly. For a finite block $\mathcal{B}$, put

$$(9.25l) \quad \mathcal{D}_{\mathcal{B}} q = \bigoplus_{n \in \mathcal{B}} \sqrt{\frac{\Lambda(n)}{\sqrt{n}}} D_n q, \quad U_{\mathcal{B}} = \mathcal{D}_{\mathcal{B}}|_{V_{M-1}}, \quad r_{\mathcal{B}} = \mathcal{D}_{\mathcal{B}} q_{M,X}^*.$$

The same minimization proves the block update

$$(9.25m) \quad \Delta_{\mathcal{B}} = \langle r_{\mathcal{B}}, (I + U_{\mathcal{B}} A^{-1} U_{\mathcal{B}}^*)^{-1} r_{\mathcal{B}} \rangle \geq \frac{\|r_{\mathcal{B}}\|^2}{1 + \|U_{\mathcal{B}}\|^2 / \lambda_{\min}(A)}.$$

Unlike a sum of atomwise floors, (9.25m) retains complementarity between prime sensors. It gives a finite interval-certifiable crossing criterion: partition the unused prime powers into finitely many blocks and verify that the sum of their lower bounds exceeds the current negative pivot.

Remark 9.7 (Why discrete Fermi levels do not force a crossing). The crossing still requires the physical theta features in (9.25i). An exact three-state reversible countermodel shows why. Let G be the path Laplacian on three vertices with both edge weights $1/8$, and let Π project onto the centered plane. Its centered eigenvalues are $1/8, 3/8$. Choose positive numbers a_n , indexed by prime powers, with $\sum a_n = 1/8$, and factor $a_n \Pi = (\Lambda(n)/\sqrt{n}) D_n^* D_n$. Every channel is then strictly observable and every finite head has discrete spectrum, while the completed eigenvalues are

$$(9.25n) \quad 0, \quad \frac{1}{4}, \quad \frac{1}{2}.$$

The last vector is an exact threshold radical, but the $1/4$ bound state never crosses. Thus reversibility, discreteness, literal positive weights, strict observability and norm convergence do not imply frontier finiteness; the missing estimate must use the actual ordinary-prime theta geometry.

9.7. Local detection and cumulative adaptive gain. The strict observability statement can be localized at the midpoint of one prime-power displacement. For $u, \eta > 0$, put $t = u/2$ and define

$$(9.25o) \quad \mathcal{J}_{u,\eta}(q, s) := \int_{-\eta}^{\eta} K(t+y) K(t-y) \{q(t+y) - q(t-y)\} \\ \times \overline{\{s(t+y) - s(t-y)\}} dy.$$

Let $\mathbf{C}_M^{\text{loc}}(u, \eta)$ be its Gram matrix in $(\phi_1, \dots, \phi_M)$.

Proposition 9.8 (Local Christoffel detection [27, 28]). *For every $u, \eta > 0$,*

$$(9.25p) \quad \mathbf{C}_M^{\text{loc}}(u, \eta) \succ 0, \quad \boxed{\kappa_M(u, \eta) = \frac{\det \mathbf{C}_M^{\text{loc}}(u, \eta)}{\det \mathbf{C}_{M-1}^{\text{loc}}(u, \eta)} > 0.}$$

Every affine residual $q = \phi_M + v$, $v \in V_{M-1}$, satisfies $\mathcal{J}_{u,\eta}(q, q) \geq \kappa_M(u, \eta)$. Hence an omitted prime power $n = p^k$ gives the adaptive lower bound

$$(9.25q) \quad \Delta_n \geq \frac{\Lambda(n)}{\sqrt{n}} \kappa_M(\log n, \eta).$$

Proof. If the local integral vanishes, strict positivity of K gives $q(t+y) = q(t-y)$ on an interval. Analytic continuation makes q invariant under reflection about t . Since q is even, composition with reflection about zero makes it u -periodic. Every vector in the finite elementary zero-mode and confluent-jet space tends to zero at $+\infty$; therefore $q = 0$. The local Gram is positive definite. Minimizing its quadratic polynomial over the coefficient of V_{M-1} gives the last Schur complement, which is the determinant ratio in (9.25p). Restricting the literal atom to the midpoint aperture and dropping the nonnegative preceding-mode penalty gives (9.25q). $\square$

The atomwise statement has an exact cumulative version. Partition the omitted prime powers into consecutive finite blocks $\mathcal{B}_1, \mathcal{B}_2, \dots$. At stage ℓ , let $A_{\ell-1} \succ 0$ be the preceding-mode block, let U_ℓ be the old-mode feature of $\mathcal{B}_\ell$, and let ρ_ℓ be its adaptive new-mode response. Set

$$(9.25r) \quad B_\ell = U_\ell A_{\ell-1}^{-1} U_\ell^*, \quad R_\ell = \|\rho_\ell\|^2, \quad C_\ell = \|A_{\ell-1}^{-1/2} U_\ell^* \rho_\ell\|^2.$$

Proposition 9.9 (Cumulative adaptive block gain). *If the initial pivot is $-\delta$, then after L blocks*

$$(9.25s) \quad \sigma_L = -\delta + \sum_{\ell=1}^L g_\ell, \quad g_\ell = \langle \rho_\ell, (I + B_\ell)^{-1} \rho_\ell \rangle \geq \frac{R_\ell^2}{R_\ell + C_\ell}.$$

Consequently a finite prefix crosses if and only if the completed exact gain satisfies $\sum_{\ell \geq 1} g_\ell > \delta$. The sufficient two-moment test is

$$(9.25t) \quad \sum_{\ell \geq 1} \frac{R_\ell^2}{R_\ell + C_\ell} > \delta.$$

Proof. The block Woodbury identity gives $\sigma_\ell - \sigma_{\ell-1} = g_\ell$, so the first identity telescopes. If ν_ℓ is the spectral measure of B_ℓ at ρ_ℓ , then

$$g_\ell = \int \frac{d\nu_\ell(s)}{1+s}, \quad R_\ell = \int d\nu_\ell(s), \quad R_\ell + C_\ell = \int (1+s) d\nu_\ell(s).$$

Cauchy–Schwarz gives $R_\ell^2 \leq g_\ell(R_\ell + C_\ell)$. Monotonicity of the partial sums proves the finite-selection assertion. $\square$

Neither strict local detection nor (9.25s) by itself forces a crossing. A nonzero completed radical has negative pivot equal to the omitted tail at every proper cutoff and acquires exactly that amount only in the limit. The radical must therefore be removed before its tail response is compared with the source deficit.

9.8. Radical-conditioned tail gain. Let $\Phi : \mathbb{C}^{M-1} \rightarrow V_{M-1}$ and $\Psi : \mathbb{C}^J \rightarrow \mathfrak{R}_J$ be coordinate maps. Realize the omitted Euler bank by a Hilbert-space feature $\mathcal{V}_X$ satisfying

$$(9.25u) \quad \|\mathcal{V}_X q\|^2 = \mathcal{T}_X(q, q), \quad U = \mathcal{V}_X \Phi, \quad v = \mathcal{V}_X \phi_M, \quad W_J = \mathcal{V}_X \Psi.$$

Strict tail positivity on the finite radical space gives $W_J^* W_J \succ 0$. Put

$$(9.25v) \quad \Pi_{J,X} = I - W_J (W_J^* W_J)^{-1} W_J^*.$$

Thus $\Pi_{J,X}$ removes precisely the tail features synthesized by a finite exact-radical correction.

Write the completed mode matrix on $V_{M-1} \oplus \text{span}\{\phi_M\}$ as H_∞ . Maximizing the finite-head form in the radical coordinate gives the exact anti-short

$$(9.25w) \quad \boxed{\hat{H}_{M,J,X} = H_\infty - \begin{pmatrix} U^* \\ v^* \end{pmatrix} \Pi_{J,X} \begin{pmatrix} U & v \end{pmatrix} = \begin{pmatrix} \hat{A} & \hat{c} \\ \hat{c}^* & \hat{h} \end{pmatrix}.}$$

Assume $\hat{A} \succ 0$, and define

$$(9.25x) \quad \hat{a} = \hat{A}^{-1}\hat{c}, \quad q_{M,J,X}^* = \phi_M - \Phi\hat{a}, \quad \hat{\sigma}_{M,J,X} = \hat{h} - \hat{c}^*\hat{A}^{-1}\hat{c} = -\delta_{M,J,X} < 0.$$

The stationary radical coefficient and full saddle residual are

$$(9.25y) \quad \zeta_{M,J,X}^* = -(W_J^*W_J)^{-1}W_J^*\mathcal{V}_X q_{M,J,X}^*, \quad \tilde{q}_{M,J,X}^* = q_{M,J,X}^* + \Psi\zeta_{M,J,X}^*.$$

In particular,

$$(9.25z) \quad \mathcal{V}_X \tilde{q}_{M,J,X}^* = \Pi_{J,X} \mathcal{V}_X q_{M,J,X}^*.$$

Theorem 9.10 (Exact radical-conditioned adaptive gain). *Set*

$$z_{M,J,X} = \Pi_{J,X} \mathcal{V}_X q_{M,J,X}^*, \quad \bar{U} = \Pi_{J,X} U.$$

Then the completed last pivot is

$$(9.25aa) \quad \boxed{\sigma_{M,\infty} = -\delta_{M,J,X} + \mathfrak{g}_{M,J,X}, \quad \mathfrak{g}_{M,J,X} = \left\langle z_{M,J,X}, (I + \bar{U}\hat{A}^{-1}\bar{U}^*)^{-1} z_{M,J,X} \right\rangle.}$$

Equivalently,

$$(9.25ab) \quad \boxed{\mathfrak{g}_{M,J,X} = \sup_{\omega \neq 0} \frac{|\langle \omega, z_{M,J,X} \rangle|^2}{\|\omega\|^2 + \|\hat{A}^{-1/2}\bar{U}^* \omega\|^2}.}$$

Finite-support tail combiners are sufficient in the strict inequality. Thus

$$(9.25ac) \quad \boxed{\sigma_{M,\infty} > 0 \iff \mathfrak{g}_{M,J,X} > \delta_{M,J,X}.}$$

Proof. The radical block in the joint finite-head matrix is $-W_J^*W_J$. Its Schur complement is (9.25w), while its normal equation gives (9.25y)–(9.25z). Adding the omitted bank back to $\hat{H}_{M,J,X}$ is therefore the positive update with old feature $\bar{U}$ and adaptive response $z_{M,J,X}$. The block Woodbury identity gives (9.25aa). The Rayleigh dual formula for the positive operator $I + \bar{U}\hat{A}^{-1}\bar{U}^*$ gives (9.25ab). Finite-support vectors are dense in the tail direct sum, and the two quadratic functionals in (9.25ab) are continuous; a strict witness may therefore be approximated by a finite-support witness. Finally (9.25ac) follows from (9.25aa). $\square$

The formula also has the joint variational form

$$(9.25ad) \quad \boxed{\mathfrak{g}_{M,J,X} = \inf_{\substack{d \in V_{M-1} \\ r \in \mathfrak{R}_J}} \left\{ \hat{\mathcal{A}}(d, d) + \|\mathcal{V}_X(q_{M,J,X}^* + d + r)\|^2 \right\},}$$

where $\hat{\mathcal{A}}$ is the positive old-mode form with matrix $\hat{A}$. This identity keeps one common radical correction and one common old-mode correction across all prime powers.

9.9. Radical-conditioned Christoffel and finite determinant exhaustion. On the elementary analytic space $V_M \oplus \mathfrak{R}_J$, every literal midpoint form (9.25o) is positive definite. Indeed, a null vector is periodic by the reflection argument. Its zero-mode component decays, whereas a nonzero largest radical derivative has a nonconstant leading polynomial in e^{2x} ; their sum cannot be periodic. Hence, if $\mathbf{C}_{M,J}^{\text{loc}}(u, \eta)$ is the local Gram in a basis of $V_M \oplus \mathfrak{R}_J$,

$$(9.25ae) \quad \kappa_{M,J}(u, \eta) = \frac{\det \mathbf{C}_{M,J}^{\text{loc}}(u, \eta)}{\det \mathbf{C}_{M-1,J}^{\text{loc}}(u, \eta)} > 0.$$

Consequently every finite omitted block $\mathcal{B}$ gives the valid but not necessarily sharp certificate

$$(9.25af) \quad \mathfrak{g}_{M,J,X} \geq \sum_{n \in \mathcal{B}} \frac{\Lambda(n)}{\sqrt{n}} \kappa_{M,J}(\log n, \eta_n).$$

The separate minima in (9.25af) can lose the joint alignment. The following augmented determinant removes that loss.

For $Y > X$, define

$$(9.25ag) \quad \mathfrak{C}_{M,J;X,Y} := \inf_{\substack{d \in V_{M-1} \\ r \in \mathfrak{R}_J}} \left\{ \widehat{\mathcal{A}}(d, d) + \sum_{\substack{n=p^k \\ X < n \leq Y}} \frac{\Lambda(n)}{\sqrt{n}} \mathcal{J}_{\log n}(q_{M,J,X}^* + d + r, q_{M,J,X}^* + d + r) \right\}.$$

In the ordered affine basis $(V_{M-1}, \mathfrak{R}_J; q_{M,J,X}^*)$, let $\mathbb{G}_n$ be the Gram matrix of $\mathcal{J}_{\log n}$, and put

$$(9.25ah) \quad \mathbb{C}_{M,J;X,Y} = \text{diag}(\widehat{\mathcal{A}}, 0_J, 0) + \sum_{\substack{n=p^k \\ X < n \leq Y}} \frac{\Lambda(n)}{\sqrt{n}} \mathbb{G}_n.$$

Let $\mathbb{C}_{M,J;X,Y}^-$ be the principal block obtained by deleting the last row and column.

Theorem 9.11 (Augmented finite determinant exhaustion). *For every Y containing at least one omitted prime power,*

$$(9.25ai) \quad \mathfrak{C}_{M,J;X,Y} = \frac{\det \mathbb{C}_{M,J;X,Y}}{\det \mathbb{C}_{M,J;X,Y}^-} > 0.$$

Moreover,

$$(9.25aj) \quad 0 < \mathfrak{C}_{M,J;X,Y} < \mathfrak{g}_{M,J,X}, \quad \mathfrak{C}_{M,J;X,Y} \nearrow \mathfrak{g}_{M,J,X},$$

and, on each fixed finite row, there are $C, c > 0$ and a polynomial Q such that

$$(9.25ak) \quad 0 \leq \mathfrak{g}_{M,J,X} - \mathfrak{C}_{M,J;X,Y} \leq CQ(Y)e^{-cY}.$$

Consequently

$$(9.25al) \quad \sigma_{M,\infty} > 0 \iff \mathfrak{C}_{M,J;X,Y} > \delta_{M,J,X} \text{ for some finite } Y.$$

Proof. A single displacement Gram is positive definite on $V_M \oplus \mathfrak{R}_J$ by the periodicity/asymptotic argument above. The positive ordinary weights therefore make both the matrix in (9.25ah) and its preceding principal block positive definite. Its last Schur complement is exactly the minimization (9.25ag), proving (9.25ai).

The objectives in (9.25ag) increase pointwise with Y and are bounded above by (9.25ad). One fixed nonempty head is coercive on the finite-dimensional correction space, so all finite minimizers lie in a common compact set. Passing to a convergent subsequence and then using monotone convergence of the nonnegative prime-power series proves that the limit is (9.25ad). If (d_∞, r_∞) minimizes the complete objective, then $q_{M,J,X}^* + d_\infty + r_\infty \neq 0$; every omitted literal atom is strictly positive on it. Every proper cutoff therefore omits a positive term, proving the strict inequality in (9.25aj). The fixed-space theta-overlap estimate bounds the remaining Gram tail by $CQ(Y)e^{-cY}$; compactness of the minimizers gives (9.25ak). Finally combine (9.25aa) and (9.25aj): a positive limiting margin is eventually larger than the vanishing tail loss, while a finite crossing implies $\mathfrak{g}_{M,J,X} > \delta_{M,J,X}$. This proves (9.25al). $\square$

The finite source equation eliminates the abstract deficit. Namely,

$$(9.25am) \quad \delta_{M,J,X} = \frac{1}{2} \|\tilde{q}_{M,J,X}^*\|_{\mu_K}^2 - \mathcal{E}_\Gamma(\tilde{q}_{M,J,X}^*) - \sum_{p^k \leq X} \frac{\log p}{p^{k/2}} \mathcal{J}_{k \log p}(\tilde{q}_{M,J,X}^*, \tilde{q}_{M,J,X}^*).$$

There is an exact cancellation which rules out a raw-response shortcut. Put

$$R_{M,J,X} = \|z_{M,J,X}\|^2, \quad \mathfrak{l}_{M,J,X} = R_{M,J,X} - \mathfrak{g}_{M,J,X} \geq 0.$$

Radical invariance of the complete form, (9.25z), and $\mathcal{A}_X = \mathcal{A}_\infty - \mathcal{T}_X$ give (9.25an)

$$R_{M,J,X} - \delta_{M,J,X} = \mathcal{A}_\infty(q_{M,J,X}^*, q_{M,J,X}^*), \quad \mathfrak{g}_{M,J,X} - \delta_{M,J,X} = \mathcal{A}_\infty(q_{M,J,X}^*, q_{M,J,X}^*) - \mathfrak{l}_{M,J,X}.$$

Thus even the prerequisite $R_{M,J,X} > \delta_{M,J,X}$ is already positivity of the completed form on the current residual. Bounds which first dominate the raw response and then pay a positive regression penalty are strictly stronger than the required statement. The augmented determinant is sharp because it retains $\mathfrak{l}_{M,J,X}$ exactly.

The same assertion admits a source-balanced bordered-determinant form. Write

$$\tau_{d+1}(Y) = \det \mathbb{C}_{M,J;X,Y}, \quad \tau_d(Y) = \det \mathbb{C}_{M,J;X,Y}^-.$$

let e_* denote the last coordinate vector, and set

$$\mathbb{P}_{M,J;X,Y} = \mathbb{C}_{M,J;X,Y} - \delta_{M,J,X} e_* e_*^*.$$

Taking the Schur complement of the preceding principal block gives the exact identity

$$\det \mathbb{P}_{M,J;X,Y} = \tau_d(Y) \{ \mathbb{C}_{M,J;X,Y} - \delta_{M,J,X} \} = \tau_{d+1}(Y) - \delta_{M,J,X} \tau_d(Y).$$

Since $\tau_d(Y) > 0$, finite crossing is therefore precisely the strict minor domination

$$\tau_{d+1}(Y) > \delta_{M,J,X} \tau_d(Y).$$

This recoding introduces neither a localization loss nor a separate effective-resistance estimate.

Thus (9.25al) is the completely finite inequality

$$(9.25ao) \quad \frac{\det \mathbb{C}_{M,J;X,Y}}{\det \mathbb{C}_{M,J;X,Y}^-} + \mathcal{E}_\Gamma(\tilde{q}_{M,J,X}^*) + \sum_{p^k \leq X} \frac{\log p}{p^{k/2}} \mathcal{J}_{k \log p}(\tilde{q}_{M,J,X}^*, \tilde{q}_{M,J,X}^*) > \frac{1}{2} \|\tilde{q}_{M,J,X}^*\|_{\mu_K}^2.$$

It retains Gamma, the scalar threshold, the exact radical correction, the old-mode adaptation, every retained ordinary prime power, and one finite omitted block in a single determinant.

9.10. Signed amplitude transfer and adaptive Krylov recovery. Fix one row (M, J, X, Y) , put $d = M - 1 + J$, and define the individual prime-power feature maps by

$$(\mathcal{D}_n q)(x) = \left(\frac{\Lambda(n)}{\sqrt{n}} K(x) K(x - \log n) \right)^{1/2} (q(x) - q(x - \log n)), \quad n = p^k.$$

Set

$$\mathcal{D}_{X,Y} = \bigoplus_{\substack{n=p^k \\ X < n \leq Y}} \mathcal{D}_n.$$

This is a square-root feature map for the finite omitted prime-power block:

$$(9.25\text{ap}) \quad \|\mathcal{D}_{X,Y} q\|^2 = \sum_{\substack{n=p^k \\ X < n \leq Y}} \frac{\Lambda(n)}{\sqrt{n}} \mathcal{J}_{\log n}(q, q).$$

Realize its observation space as $L^2(\Omega_Y, \nu_Y)$, where Ω_Y is the disjoint union of the channel copies of $\mathbb{R}$. The one-particle exterior space is

$$\mathcal{H}_Y = \mathbb{C}^{M-1} \oplus L^2(\Omega_Y, \nu_Y) = L^2(\tilde{\Omega}_Y, \tilde{\nu}_Y),$$

where $\tilde{\Omega}_Y$ adjoins $M - 1$ counting-measure atoms. We use the normalized exterior convention

$$\|\Phi\|_{[r]}^2 = \frac{1}{r!} \int_{\tilde{\Omega}_Y^r} |\Phi|^2 d\tilde{\nu}_Y^r.$$

Abbreviate

$$A = \hat{A}, \quad U = \mathcal{D}_{X,Y} \Phi, \quad W = \mathcal{D}_{X,Y} \Psi, \quad v = \mathcal{D}_{X,Y} q_{M,J,X}^*.$$

The radical basis and finite observability give $W^* W \succ 0$. Set

$$(9.25\text{aq}) \quad \Pi_Y = I - W(W^* W)^{-1} W^*, \quad z = \Pi_Y v, \quad B = \Pi_Y U A^{-1} U^* \Pi_Y.$$

The finite adaptive gain is then

$$(9.25\text{ar}) \quad G_Y := \mathfrak{C}_{M,J,X,Y} = \langle z, (I + B)^{-1} z \rangle = \frac{\tau_{d+1}(Y)}{\tau_d(Y)}.$$

Theorem 9.12 (Exact signed amplitude transfer). *Let $\psi \in \ker W^*$ and $\beta_\psi = \langle \psi, v \rangle \neq 0$. Define*

$$(9.25\text{as}) \quad g_\psi = (-A^{-1/2} U^* \psi, \psi), \quad h_\psi = \frac{(-1)^d \sqrt{\delta_{M,J,X}}}{\beta_\psi} g_\psi.$$

If Δ_d and Δ_{d+1} are the nuisance and augmented Andreief amplitudes of the column system

$$(A^{1/2} e_i, U e_i), \quad (0, W e_a), \quad (0, v),$$

then

$$(9.25\text{at}) \quad \int \overline{h_\psi(\omega)} \Delta_{d+1}(\omega, \omega) d\tilde{\nu}_Y(\omega) = \sqrt{\delta_{M,J,X}} \Delta_d(\omega).$$

The associated exterior interior-product operator has norm

$$(9.25\text{au}) \quad \|T_{h_\psi}\|_{\text{op}}^2 = \delta_{M,J,X} \frac{\|\psi\|^2 + \|A^{-1/2} U^* \psi\|^2}{|\langle \psi, v \rangle|^2}.$$

Proof. The first component of g_ψ cancels every old-mode column, while $W^*\psi = 0$ cancels every radical column. Only the new-mode cofactor survives the Laplace expansion, giving (9.25at). Contraction by a one-particle vector on an exterior power has norm $\|h_\psi\|$, which gives (9.25au). $\square$

If $z = 0$, set $G_Y = Q_k = 0$ for all k ; no negative pivot can cross at this cutoff. Assume henceforth that $z \neq 0$. The choice $\psi = z$ gives the matched-filter certificate

$$(9.25av) \quad Q_1 = \frac{m_0^2}{m_0 + m_1}, \quad m_j = \langle z, B^j z \rangle.$$

It is sufficient, but it need not detect a crossing already present in G_Y .

For $k \geq 1$, define

$$(9.25aw) \quad H_k = [m_{i+j} + m_{i+j+1}]_{i,j=0}^{k-1}, \quad b_k = (m_0, \dots, m_{k-1})^\top, \quad Q_k = b_k^* H_k^\dagger b_k.$$

Theorem 9.13 (Adaptive Krylov hierarchy). *One has*

$$(9.25ax) \quad Q_k = \sup_{0 \neq \psi \in \mathcal{K}_k} \frac{|\langle \psi, z \rangle|^2}{\langle \psi, (I+B)\psi \rangle}, \quad Q_1 \leq Q_2 \leq \dots \leq G_Y.$$

If the spectral measure of B seen from z has r support points, then $Q_r = G_Y$.

Proof. Writing $\psi = \sum_{j < k} c_j B^j z$ converts the quotient into $|c^* b_k|^2 / (c^* H_k c)$. The finite Riesz formula gives Q_k , and nestedness gives monotonicity. If $c \in \ker H_k$, its Krylov vector is zero, hence $c^* b_k = 0$; thus $b_k \in \text{ran } H_k$ and the pseudoinverse formula is well-defined. On an r -point support, $(1+t)^{-1}$ has a polynomial interpolant of degree at most $r-1$, so $(I+B)^{-1}z \in \mathcal{K}_r$. $\square$

9.11. Second-level spectral dispersion. Put

$$(9.25ay) \quad N_2 = m_0 m_2 - m_1^2, \quad D_2 = (m_0 + m_1)(m_2 + m_3) - (m_1 + m_2)^2.$$

When $D_2 > 0$, direct inversion of H_2 gives

$$(9.25az) \quad Q_2 = Q_1 + \frac{N_2^2}{(m_0 + m_1)D_2}.$$

If $D_2 = 0$, the spectral measure seen from z has one support point and $Q_2 = Q_1 = G_Y$.

Let $d\rho(t) = d\langle z, E_B(t)z \rangle$. Then

$$(9.25ba) \quad N_2 = \frac{1}{2} \iint (s-t)^2 d\rho(s) d\rho(t),$$

$$(9.25bb) \quad D_2 = \frac{1}{2} \iint (s-t)^2 (1+s)(1+t) d\rho(s) d\rho(t).$$

Equivalently,

$$(9.25bb) \quad N_2 = \|z \wedge Bz\|^2, \quad D_2 = \|(I+B)^{1/2}z \wedge (I+B)^{1/2}Bz\|^2.$$

Thus, when $Q_1 < \delta_{M,J,X}$,

$$(9.25bc) \quad N_2^2 > (\delta_{M,J,X} - Q_1)(m_0 + m_1)D_2$$

is exactly equivalent to $Q_2 > \delta_{M,J,X}$.

For any two orthonormal observation packets ϕ_0, ϕ_1 , put

$$a_i = \langle \phi_i, z \rangle, \quad b_i = \langle \phi_i, Bz \rangle.$$

The corresponding exterior coordinate gives

$$(9.25bd) \quad N_2 \geq |a_0 b_1 - a_1 b_0|^2.$$

Since $D_2 \leq (1 + \|B\|)^2 N_2$, the signed two-packet condition

$$(9.25be) \quad |a_0 b_1 - a_1 b_0|^2 > (\delta_{M,J,X} - Q_1)(m_0 + m_1)(1 + \|B\|)^2$$

implies the finite crossing.

The literal ordinary-prime form uses one common radical and old-mode regression. Put

$$\begin{aligned} \zeta^* &= -(W^* W)^{-1} W^* v, & Z &= -(W^* W)^{-1} W^* U, \\ \tilde{q} &= q_{M,J,X}^* + \Psi \zeta^*, & \tilde{\Phi} &= \Phi + \Psi Z, \end{aligned}$$

and, in the individual prime-power channels,

$$(9.25bf) \quad x_n = \mathcal{D}_n \tilde{q}, \quad F_n = \mathcal{D}_n \tilde{\Phi} A^{-1/2}, \quad \xi = \sum_{\substack{n=p^k \\ X < n \leq Y}} F_n^* x_n.$$

Then

$$(9.25bg) \quad N_2 = \frac{1}{2} \sum_{\substack{n=p^k, \ell=q^a \\ X < n, \ell \leq Y}} \iint |x_n(s)(F_\ell \xi)(t) - (F_n \xi)(s)x_\ell(t)|^2 ds dt.$$

All von Mangoldt weights and theta phases therefore remain coupled until the final modulus square.

Corollary 9.14 (Exact hierarchy formulation of the frontier). *For every fixed negative anti-shortened row,*

$$(9.25bh) \quad \mathfrak{g}_{M,J,X} > \delta_{M,J,X} \iff \exists Y < \infty, \exists k < \infty : Q_k(Y) > \delta_{M,J,X}.$$

Proof. The forward implication follows from augmented determinant exhaustion, followed by finite Krylov exactness. Conversely, $Q_k(Y) \leq G_Y \leq \mathfrak{g}_{M,J,X}$. $\square$

Remark 9.15 (Level two is a sufficient gate). The depth in Corollary 9.14 may depend on the finite row. The dispersion inequality (9.25bc) is the first explicit strengthening of MF, not a replacement for the exact surplus. For example,

$$B = \text{diag}(1, 2, 3), \quad z = \sqrt{12/13}(1, 1, 1), \quad \delta = 649/650$$

gives $G_Y = 1 > \delta$, but $Q_2 = 648/650 < \delta$. Failure at level two is therefore inconclusive. If $G_Y > \delta$, finite Krylov exactness guarantees that some higher row-dependent level detects the crossing.

Problem 9.16 (Physical strict surplus). For every negative anti-shortened row required in the cofinal exhaustion, prove, with the literal theta translations and ordinary von Mangoldt weights,

$$(9.25bi) \quad \mathfrak{g}_{M,J,X} > \delta_{M,J,X}.$$

By Theorem 9.11, this is equivalent to one finite inequality (9.25ao), and by Corollary 9.14 to one finite Krylov certificate of row-dependent depth. No localization, atomwise minimization, or infinite-selection loss remains in this formulation.

The mean-periodic constraint must be applied before the finite radical correction. Indeed, if $q \in (\mathbf{1} \oplus \mathcal{R})^\perp$ and $r \in \mathfrak{R}_J$, then

$$(h(q+r)) * K = 0 \iff r = 0,$$

because the mean-periodic complement intersects the complete radical only at zero. Thus $q_{M,J,X}^* + d$ is mean-periodic for $d \in V_{M-1}$, whereas a joint saddle vector with a nonzero radical correction is not. The correction must remain in the determinant, but the convolution equation may not be imposed on it.

9.12. Heat-core exhaustion of the radical complement. The elementary zero modes provide the explicit orbit coordinate used in the finite calculations above. Their completeness in the weighted form topology is a separate spectral-synthesis statement. It is not needed for the global passage. We now construct a form core directly from the closed ordinary-prime–Gamma form.

Let

$$(9.25bj) \quad \mathcal{C} = (\mathbf{1} \oplus \mathcal{R})^\perp, \quad \mathcal{Q}_{\mathcal{C}} = D(\mathcal{E}_K) \cap \mathcal{C}.$$

The complete form is closed. Indeed, on its maximal domain it is the squared norm of the difference operator

$$(9.25bk) \quad \mathcal{D}q = \left(\sqrt{g(u)K(x)K(x-u)}\{q(x) - q(x-u)\} \right)_{(u,x) \in (0,\infty) \times \mathbb{R}} \\ \oplus \left(\sqrt{\frac{\Lambda(n)}{\sqrt{n}}} K(x)K(x - \log n)\{q(x) - q(x - \log n)\} \right)_{n=p^a},$$

from $L_{\text{even}}^2(\mu_K)$ to the corresponding Hilbert direct sum. If $q_\ell \rightarrow q$ and $\mathcal{D}q_\ell \rightarrow Y$, one subsequence converges almost everywhere with respect to Lebesgue measure because $d\mu_K$ has a strictly positive density. Fubini gives simultaneous convergence at $x - u$, and an almost-everywhere subsequence in the disjoint-union output space identifies $Y = \mathcal{D}q$. Thus $\mathcal{D}$ is closed. Moreover $C_{c,\text{even}}^\infty(\mathbb{R}) \subset D(\mathcal{D})$: near $u = 0$, $g(u) = (2u)^{-1} + O(1)$ and the squared difference is $O(u^2)$, while for large u the theta overlap is $O_q(e^{-c_q e^{2u}})$. At $u = \log n$ this is summable against $\Lambda(n)/\sqrt{n}$. Hence the form is densely defined and closed. Let $L \geq 0$ be its associated self-adjoint operator.

Equation (9.15), interpreted by the representation theorem, gives the strong domain identities

$$(9.25bl) \quad L\mathbf{1} = 0, \quad r_j \in D(L), \quad Lr_j = \frac{1}{2}r_j.$$

Therefore $\mathbf{1} \oplus \mathcal{R}$ and $\mathcal{C}$ reduce L . Put

$$(9.25bm) \quad S_{\mathcal{C}} = L|_{\mathcal{C}} + \frac{1}{2}I \geq \frac{1}{2}I.$$

Choose a sequence $(g_j)_{j \geq 1} \subset \mathcal{C}$ whose algebraic span is Hilbert-dense and define

$$(9.25bn) \quad V_M^{\text{heat}} = \text{span}\{e^{-S_{\mathcal{C}}/k} g_j : 1 \leq j, k \leq M\}.$$

Theorem 9.17 (Heat-core Galerkin exhaustion). *The spaces V_M^{heat} are nested finite-dimensional subspaces of $\mathcal{Q}_{\mathcal{C}}$, every one of their vectors belongs to $\cap_{m \geq 0} D(S_{\mathcal{C}}^m)$, and*

$$(9.25bo) \quad \overline{\bigcup_M V_M^{\text{heat}}}^{(\frac{1}{2}\|\cdot\|_{\mu_K}^2 + \mathcal{E}_K)^{1/2}} = \mathcal{Q}_{\mathcal{C}}.$$

For every fixed finite-dimensional $E \subset D(\mathcal{E}_K)$, the omitted Euler tail satisfies

$$(9.25bp) \quad \left\| N_E^{-1/2}(\mathcal{T}_X)_E N_E^{-1/2} \right\| \longrightarrow 0.$$

Finally, if a finite omitted block contains two distinct primes p, r , its Gram form is positive definite on every finite centered space $E \oplus \mathfrak{R}_J$ with $E \subset \mathcal{C}$.

Proof. Spectral calculus gives, for $t > 0$,

$$(9.25bq) \quad \|S_{\mathcal{C}}^m e^{-tS_{\mathcal{C}}}\| \leq \left(\frac{m}{et}\right)^m, \quad \|S_{\mathcal{C}}^{1/2} e^{-tS_{\mathcal{C}}}\| \leq (2et)^{-1/2}.$$

For $q \in D(S_{\mathcal{C}}^{1/2})$, dominated convergence in its spectral measure gives

$$(9.25br) \quad \|S_{\mathcal{C}}^{1/2}(e^{-tS_{\mathcal{C}}} - I)q\| \longrightarrow 0.$$

Given $\varepsilon > 0$, first choose k so that (9.25br), with $t = 1/k$, is smaller than $\varepsilon/2$. Approximate q in the Hilbert norm by a finite linear combination g of the g_j , with $\|q - g\| < (\varepsilon/2)\sqrt{2et}$. Equations (9.25bq)–(9.25br) then give

$$\|S_{\mathcal{C}}^{1/2}(q - e^{-tS_{\mathcal{C}}}g)\| < \varepsilon,$$

which proves (9.25bo).

For (9.25bp), choose a basis $e_1, \dots, e_d$ of E . Positivity and monotone convergence give $\mathcal{T}_X(e_i, e_i) \downarrow 0$, while Cauchy–Schwarz for the tail form gives

$$|\mathcal{T}_X(e_i, e_j)|^2 \leq \mathcal{T}_X(e_i, e_i)\mathcal{T}_X(e_j, e_j).$$

Thus every entry, and hence the normalized finite matrix, tends to zero.

For the last assertion, vanishing of the two prime atoms would give the two periods $\log p$ and $\log r$. Their quotient is irrational, since a rational relation would imply $p^a = r^b$. Strong continuity of translations in L_{loc}^2 then makes every real number a period, so the vector is constant almost everywhere. Centeredness makes it zero. The finite Gram matrix is therefore positive definite. $\square$

The hybrid spaces

$$(9.25bs) \quad V_M^{\text{hyb}} = V_M^{\text{heat}} + V_M$$

are also a form core. They retain every explicit zero-orbit diagnostic used above, without asserting that the canonical zero modes alone are dense. One-channel observability and the local scalar Christoffel bounds remain statements about the canonical summand. In the global closure, finite tail blocks must contain two distinct primes, and the physical surplus must be proved on the heat or hybrid rows.

9.13. The compensated frontier and closure. The strict frontier (9.12) excludes a completed threshold null vector and can therefore be stronger than mere nonnegativity. The exact vanishing-tolerance formulation is the following. Choose

$$(9.26) \quad M \longmapsto (J(M), X(M), \varepsilon_M), \quad J(M), X(M) \rightarrow \infty, \quad \varepsilon_M \downarrow 0,$$

and require

$$(9.27) \quad n_- \left[\mathbf{A}_{M, J(M), X(M)} + \varepsilon_M \begin{pmatrix} \mathbf{N}_M & 0 \\ 0 & 0 \end{pmatrix} \right] = J(M).$$

Equivalently,

$$(9.28) \quad \boxed{\mathbf{T}^{VV} - \mathbf{T}^{VR}(\mathbf{T}^{RR})^{-1}\mathbf{T}^{RV} \preceq \mathbf{H}_M + \varepsilon_M \mathbf{N}_M.}$$

On the canonical zero-mode summand the cutoff part of (9.28) has the explicit bound $C_M e^{-cX(M)} \mathbf{N}_M$. On every heat or hybrid row, Theorem 9.17 gives the qualitative bound needed for the diagonal schedule: after the finite row is fixed, $X(M)$ may be enlarged until the normalized tail is at most ε_M . The remaining content is the joint signed prime–Gamma floor.

Theorem 9.18 (Cofinal inertia closure). *Let V_M be either the heat spaces (9.25bn) or the hybrid spaces (9.25bs), and assume that (9.27) holds. Then*

$$(9.29) \quad \mathcal{A}_\infty(q, q) \geq 0$$

on that complement. Under the moving/model identification this gives the source-side Mosco lower bound, and hence RH.

Proof. First synchronize the inertia and tail cutoffs. If (9.27) holds at $\widehat{X}(M)$, replace it by any

$$X(M) \geq \max\{\widehat{X}(M), X_{\text{tail}}(M, J(M), \varepsilon_M)\}.$$

Increasing the cutoff adds a positive Gram matrix, so the negative index cannot increase. It cannot fall below $J(M)$, because the shifted form on every nonzero $r \in \mathfrak{R}_{J(M)}$ is

$$\mathcal{A}_{X(M)}(r, r) = -\mathcal{T}_{X(M)}(r, r) < 0;$$

the remaining tail contains two distinct primes and Theorem 9.17 applies. Thus (9.27) persists at the synchronized cutoff.

Now fix $q \in V_m$. For every $M \geq m$, (9.27), the Schur-complement formula, and the inclusion of q in the larger mode block give

$$\mathcal{A}_\infty(q, q) \geq -\varepsilon_M \|q\|_{\mu_K}^2.$$

Let $M \rightarrow \infty$. Then $\varepsilon_M \rightarrow 0$, so $\mathcal{A}_\infty(q, q) \geq 0$ on $\cup_M V_M$. Theorem 9.17 and continuity in the closed form norm extend the inequality to the complete centered radical complement. The radical has value zero by (9.15), while constants have zero value in $\mathcal{E}_K - \frac{1}{2} \text{Var}_{\mu_K}$. Hence the complete centered, equivalently constant-quotiented, Weil form is nonnegative. Theorem 1.1 gives RH. Equivalently, in the CCM branch coordinate, Theorems 8.10 and 8.12 remove the already controlled residual pieces and (9.27) supplies the source liminf in Problem 8.3. $\square$

Component	Status in this proof chain
CCM finite self-adjoint quotient	Published finite package.
Connes prolate critical-line model	Published finite/model package.
Suzuki localized operator	Published localized realization.
Even Rayleigh residual	Proved: $ \mathcal{R}_L^+ \ll \lambda^7 d_4$.
Direct cross residual	Proved: $\ b_L^+\ \ll \lambda^4 \sqrt{(1+L)d_4} \rightarrow 0$.
Literal Euler-head approximation	Explicit canonical rate proved by (9.7)–(9.8); qualitative normalized convergence on every heat/hybrid row proved by Theorem 9.17.
Finite radical inertia algebra	Proved by (9.16)–(9.22).
Large-prime channel geometry	Proved by (9.23)–(9.25).
Finite one-channel observability	Proved on the canonical zero-mode rows by Theorem 9.5; finite rank and aliasing are removed there. General heat rows use two-prime blocks.
Completed scalar innovation	Exact recursion proved by Theorem 9.6.
Prime-by-prime staircase update	Exact positive Kalman–Schur increment proved in (9.25j)–(9.25k), with the literal $\Lambda(p^k)$ weights.
Local and cumulative gain	Midpoint Christoffel detection and the exact block telescope are proved in Proposition 9.8 and Proposition 9.9.
Radical-conditioned gain	The exact projected residual, resolvent gain, and matched-filter criterion are proved in Theorem 9.10.
Augmented finite determinant	Strict positivity, determinant formula, monotone exhaustion, and finite selection are proved in Theorem 9.11.
Signed amplitude transfer	The exact exterior interior-product identity and its norm are proved in Theorem 9.12.
Adaptive Krylov recovery	Monotonicity and finite-row exactness are proved in Theorem 9.13.
Second-level dispersion gate	Formula (9.25az), its spectral wedge identity, and the sufficient test (9.25bc) are proved; the test is not asserted uniformly for every physical row.
Physical strict surplus (9.25bi)	Remaining force-bearing arithmetic inequality; it must hold on the heat or hybrid form-core rows, with no localization or infinite-selection loss.
Joint compensated frontier (9.27)	Follows row by row from the strict surplus; its vanishing-tolerance form permits threshold equality.
Prime–Gamma form-core exhaustion	Proved by Theorem 9.17; canonical zero-mode-only synthesis is not asserted or required.

Remark 9.19 (Strict and compensated targets). The strict statement requested by the finite-energy-level picture is

$$(9.30) \quad \boxed{X_*(M, J(M)) < \infty \quad \text{for every } M.}$$

By Theorem 9.2, its force-bearing content is a strict completed finite-mode gap. The compensated statement (9.27) is the threshold-stable version needed for RH. A proof of (9.30) on a form-core exhaustion proves (9.27) immediately; a proof of (9.27) need not exclude additional threshold null states at finite M .

10. TATE BOUNDARY GEOMETRY AND THE GLOBAL POLARIZATION AUDIT

The finite-inertia analysis above isolates the missing sign but does not explain where that sign could come from. This section records a second, source-level construction. It starts from finite root graphs at the ordinary primes, constructs the Gamma and polar determinant at infinity, and joins them before completion. The construction produces the correct local masses, parity, boundary cancellation and scale covariance. It also permits a decisive audit of the ordinary Hilbert completion.

The distinction maintained throughout is

$$(10.1) \quad \boxed{\text{positive local/source polarization} \not\Rightarrow \text{positive faithful polarization of } H_{\text{CCM}}^1}$$

The left side will be constructed below. The most natural absolutely continuous realization of the arrow in (10.1) will be disproved.

10.1. Finite root geometry and the literal von Mangoldt mass. For $N \geq 2$, let $R_N = \mathbb{Z}/N\mathbb{Z}$, let T_N be translation by one on $\ell^2(R_N)$, and put

$$(10.2) \quad \Delta_N = 2I - T_N - T_N^*.$$

Its form is the positive graph energy

$$(10.3) \quad \langle f, \Delta_N f \rangle = \sum_{j \in R_N} |f(j+1) - f(j)|^2.$$

The kernel consists of the constants. We write $\det' \Delta_N$ for the product of the nonzero eigenvalues.

Theorem 10.1 (Root torsion–incidence identity). *For every $N \geq 2$,*

$$(10.4) \quad \det' \Delta_N = N^2.$$

Fix $m \geq 2$, a prime p , and $k \geq 1$. Then

$$(10.5) \quad \frac{1}{2} \log \frac{\det' \Delta_{mp^k}}{\det' \Delta_{mp^{k-1}}} = \log p.$$

If $H_{m,k} = p^k R_m \subset R_{mp^k}$, and

$$\omega_{mp^k} = (mp^k)^{-1/2} \mathbf{1}_{R_{mp^k}}, \quad \eta_{m,k} = m^{-1/2} \mathbf{1}_{H_{m,k}},$$

then

$$(10.6) \quad \langle \omega_{mp^k}, \eta_{m,k} \rangle = p^{-k/2}.$$

Consequently

$$(10.7) \quad \boxed{\left(\frac{1}{2} \log \frac{\det' \Delta_{mp^k}}{\det' \Delta_{mp^{k-1}}} \right) \langle \omega_{mp^k}, \eta_{m,k} \rangle = \frac{\Lambda(p^k)}{\sqrt{p^k}}.}$$

Proof. The Fourier characters of R_N diagonalize Δ_N , with eigenvalues $|1 - e^{2\pi ij/N}|^2$. Hence

$$\det \Delta_N = \left| \prod_{j=1}^{N-1} (1 - e^{2\pi ij/N}) \right|^2 = N^2,$$

which gives (10.4) and (10.5). The subgroup $H_{m,k}$ has m elements, so the normalized overlap is $m/(\sqrt{mp^k}\sqrt{m}) = p^{-k/2}$. Multiplying by (10.5) proves (10.7). $\square$

The two factors in (10.7) have different meanings. The logarithm is a relative determinant increment along the root tower, whereas $p^{-k/2}$ is a normalized incidence number. The distinction prevents the incorrect replacement of the weight of the layer p^k by $k \log p$.

The finite root levels are polarized before any global sign is considered. On $H_N^0 = \mathbf{1}^\perp \subset \ell^2(R_N)$, set

$$\begin{aligned} J_N(x, y) &= (-y, x), \\ g_N((x, y), (u, v)) &= \langle \Delta_N x, u \rangle + \langle \Delta_N y, v \rangle, \\ \Omega_N(z, w) &= -g_N(z, J_N w). \end{aligned} \tag{10.8}$$

Then g_N is positive definite, $J_N^2 = -I$, and Ω_N is alternating and nondegenerate. If $M \mid N$, normalized pullback along $R_N \rightarrow R_M$ is a polarized isometry and the determinant on the new-root complement is $(N/M)^2$. Thus (10.8) is functorial under the root coverings which produce (10.5).

10.2. Poisson holonomy, heat lift and local unitary triviality. The coefficient $p^{-k/2}$ also has an exact positive realization as a unitary moment. For $0 < r < 1$, let

$$d\mu_r(\theta) = \frac{1 - r^2}{1 - 2r \cos \theta + r^2} \frac{d\theta}{2\pi}. \tag{10.8a}$$

The Fourier expansion of the Poisson kernel gives

$$\int_{-\pi}^{\pi} e^{ik\theta} d\mu_r(\theta) = r^{|k|}. \tag{10.8b}$$

Taking $r = p^{-1/2}$ realizes every winding of the prime tower without analytic continuation.

Let $A_{p,\theta}$ be the nonnegative Laplacian on the circle of length $\ell_p = \log p$, with boundary condition $f(x + \ell_p) = e^{i\theta} f(x)$. The method-of-images kernel is

$$K_{p,\theta,t}(x, y) = \frac{1}{\sqrt{4\pi t}} \sum_{k \in \mathbb{Z}} e^{-(x-y+k\ell_p)^2/(4t)} e^{ik\theta}. \tag{10.8c}$$

It satisfies the exact heat composition law before any winding sector is removed. Integrating its trace against (10.9) yields

$$\tau_p(e^{-tA_p}) = \frac{\ell_p}{\sqrt{4\pi t}} \left(1 + 2 \sum_{k \geq 1} p^{-k/2} e^{-(k\ell_p)^2/(4t)} \right). \tag{10.8d}$$

The nonzero-winding relative trace is therefore exactly

$$\tau_p(e^{-tA_p}) - \frac{\log p}{\sqrt{4\pi t}} = \frac{\log p}{\sqrt{\pi t}} \sum_{k \geq 1} p^{-k/2} e^{-(k \log p)^2/(4t)}. \tag{10.8e}$$

This is the complete Gaussian prime tower in the heat explicit formula.

The local realization has a rigidity which is important globally. The Poisson density is the modulus square of the outer function

$$(10.8f) \quad a_r(e^{i\theta}) = \frac{\sqrt{1-r^2}}{1-re^{i\theta}}.$$

Multiplication by a_r is a unitary map $L^2(\mathbb{T}, \mu_r) \rightarrow L^2(\mathbb{T}, d\theta/2\pi)$ and commutes with every decomposable fiber operator. Hence the weighted and unweighted local Laplacians are unitarily equivalent. The prime changes the normal weight used to read the heat diagonal; it does not create a new local spectrum. Moreover the winding-zero sector cannot be deleted while preserving composition, because two nonzero windings can add to zero. Thus the finite quantity in (10.13) is a relative semifinite trace, not the ordinary trace of a reduced semigroup.

There is an equivalent precision-space formulation. On $\ell^2(\mathbb{N}_0)$, let

$$(K_p x)_h = \sum_{k \geq 0} p^{-|h-k|/2} x_k.$$

With $a = p^{-1/2}$, define

$$(10.8g) \quad (B_a x)_0 = x_0, \quad (B_a x)_{h+1} = \frac{x_{h+1} - ax_h}{\sqrt{1-a^2}}.$$

Direct triangular inversion gives

$$(10.8h) \quad \boxed{K_p^{-1} = B_a^* B_a, \quad \langle K_p^{-1} x, x \rangle = |x_0|^2 + \frac{1}{1-p^{-1}} \sum_{h \geq 0} |x_{h+1} - p^{-1/2} x_h|^2.}$$

Thus the Poisson coefficient is also the Green kernel of a positive local first-order arithmetic connection. The local positivity is exact, while $\sum_p p^{-1} = \infty$ prevents the infinite product of these changes of coordinates from being a bounded perturbation of a reference product. The matched Gamma–Euler boundary must therefore be formed before the global norm or trace is taken.

10.3. The Gamma spin and the polar determinant. On $\mathcal{H}_\Gamma = \ell^2(\mathbb{N}_0)$, define the positive operator

$$(10.9) \quad N_\Gamma e_m = \left(2m + \frac{1}{2}\right) e_m.$$

Its heat trace is

$$(10.10) \quad \text{Tr } e^{-uN_\Gamma} = \frac{e^{-u/2}}{1 - e^{-2u}}.$$

Zeta regularization of $N_\Gamma + s - 1/2$ gives the archimedean determinant directly from this source operator.

Theorem 10.2 (Gamma–polar determinant). *For $\Re s > 0$, with meromorphic continuation thereafter,*

$$(10.11) \quad \det_\zeta(N_\Gamma + s - 1/2) = \frac{\sqrt{2\pi} 2^{1/2-s/2}}{\Gamma(s/2)}.$$

Let N_{triv} have eigenvalues $-1/2$ and $1/2$. Then

$$(10.12) \quad \boxed{\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2) = \sqrt{\pi}(2\pi)^{-s/2} \frac{\det((s-1/2)I - N_{\text{triv}})}{\det_\zeta(N_\Gamma + s - 1/2)}}.$$

Proof. The spectral zeta function is

$$\mathrm{Tr}(N_\Gamma + s - 1/2)^{-w} = 2^{-w} \zeta_H(w, s/2).$$

Use $\zeta_H(0, a) = 1/2 - a$ and $\zeta'_H(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi)$ to obtain (10.11). Since $\det((s - 1/2)I - N_{\mathrm{triv}}) = s(s - 1)$, substitution gives (10.12). $\square$

The positive degree-one Gamma page is the real double of the form domain of $N_\Gamma^{1/2}$, with metric and complex structure

$$(10.13) \quad \begin{aligned} g_\Gamma((x, y), (u, v)) &= \langle N_\Gamma^{1/2} x, N_\Gamma^{1/2} u \rangle + \langle N_\Gamma^{1/2} y, N_\Gamma^{1/2} v \rangle, \\ J_\Gamma(x, y) &= (-y, x). \end{aligned}$$

The polar plane remains in degrees zero and two. It is not inserted as a positive degree-one summand. This preserves the Lefschetz parity $H^0 - H^1 + H^2$, which is responsible for the sign of the prime terms in the explicit formula and is central to the geometric interpretation in §10.9.

10.4. Matched boundary cancellation and the full finite part. Let $\mathcal{K}$ denote the common Cauchy coefficient space. For each prime power set

$$(10.14) \quad w_{p,k} = \frac{\log p}{p^k}.$$

The restricted-product boundary is decomposed into a common component $F \in \mathcal{K}$ and local deviations $r = (r_{p,k})$, where

$$(10.15) \quad \sum_{p,k} w_{p,k} \|r_{p,k}\| + \left(\sum_{p,k} w_{p,k} \|r_{p,k}\|^2 \right)^{1/2} < \infty.$$

Thus the local row is $F_{p,k} = F + r_{p,k}$. For $s > 1/2$, put

$$(10.16) \quad C_s = \sum_p (\log p) p^{-2s}, \quad \kappa_\infty = \gamma + \sum_p \sum_{k \geq 2} w_{p,k}.$$

The ordinary prime number theorem, in its logarithmic-derivative form at the pole, gives the normalized finite-part identity

$$(10.17) \quad \mathrm{FP}_{s \downarrow 1/2} C_s = -\kappa_\infty.$$

For $x = (F, r)$, $y = (G, z)$, consider the jointly regularized form

$$(10.18) \quad \begin{aligned} \mathfrak{Q}_s(x, y) &= \sum_p (\log p) p^{-2s} \langle F + r_{p,1}, G + z_{p,1} \rangle \\ &\quad + \gamma \langle F, G \rangle + \sum_p \sum_{k \geq 2} w_{p,k} \langle F + r_{p,k}, G + z_{p,k} \rangle. \end{aligned}$$

Theorem 10.3 (Exact matched finite part). *The finite part of (10.18) exists and equals*

$$(10.19) \quad \boxed{\mathrm{FP}_{s \downarrow 1/2} \mathfrak{Q}_s(x, y) = \sum_{p,k} w_{p,k} (\langle r_{p,k}, G \rangle + \langle F, z_{p,k} \rangle + \langle r_{p,k}, z_{p,k} \rangle)}.$$

The series is absolutely convergent and defines a continuous form on the space in (10.15). In particular the common generic diagonal has zero finite part.

Proof. After expanding (10.18), the coefficient of $\langle F, G \rangle$ is $C_s + \kappa_\infty$, whose finite part is zero by (10.17). Every residual coefficient converges to $w_{p,k}$. The first summand in (10.15) controls the common-residual terms and weighted Cauchy-Schwarz controls the residual-residual term, so dominated convergence gives (10.19). $\square$

Theorem 10.3 is stronger than cancellation of two scalar divergences. It retains every independent local deviation with its literal von Mangoldt weight. Hence Gamma, the pole and the prime layers are joined before taking an absolute square or a quotient.

10.5. Covariant mixing of all prime-power charges. The local row of charge $\ell_{p,k} = k \log p$ has normalized scale generator $A - \ell_{p,k}$. Adding rows with different charges without transporting them is not equivariant. The required transport is explicit. In the spectral realization

$$(10.20) \quad \mathcal{K}_C = L^2(\mathbb{R}, w_C(\gamma) d\gamma), \quad w_C(\gamma) = \frac{1}{2\pi(\gamma^2 + 1/4)}, \quad AF(\gamma) = \gamma F(\gamma),$$

define

$$(10.21) \quad (S_\ell F)(\gamma) = \left(\frac{w_C(\gamma + \ell)}{w_C(\gamma)} \right)^{1/2} F(\gamma + \ell).$$

Then $(S_\ell)_{\ell \in \mathbb{R}}$ is a strongly continuous unitary group and

$$(10.22) \quad S_\ell(A - \ell I) = AS_\ell, \quad S_\ell^* f(A) S_\ell = f(A - \ell I).$$

For $I_X = \{(p, k) : p^k \leq X\}$, put

$$(10.23) \quad c_p = \frac{2\pi}{\log p}, \quad \alpha_{p,k} = \sqrt{\frac{\log p}{c_p}} p^{-k/2}.$$

The covariant Tate boundary is

$$(10.24) \quad \begin{aligned} R_X^{\text{cov}} v &= \sum_{p,k} \alpha_{p,k} S_{\ell_{p,k}} y_{p,k}, \\ R_X^{\text{cov}} J_X v &= \sum_{p,k} c_p \alpha_{p,k} S_{\ell_{p,k}} x_{p,k}. \end{aligned}$$

It intertwines every charged source action with the one action generated by A . Its adjoint generic plane has Gram mass

$$(10.25) \quad C_X I, \quad C_X = \sum_{p^k \leq X} \frac{\log p}{p^k}.$$

Let the complete Gamma compliance be

$$(10.26) \quad K_\Gamma = \kappa_\infty I + m_\Gamma(A) \text{succ} q \kappa_\infty I.$$

Shorting only the common archimedean boundary gives the positive metric

$$(10.27) \quad \begin{aligned} g_X^{\text{cov}}(v, w) &= g_X(v, w) \\ &+ \langle K_\Gamma^{-1} R_X^{\text{cov}} J_X v, R_X^{\text{cov}} J_X w \rangle \\ &+ \langle K_\Gamma^{-1} R_X^{\text{cov}} v, R_X^{\text{cov}} w \rangle. \end{aligned}$$

The diagonal row is

$$(10.28) \quad S_{\ell_{p,k}}^* K_\Gamma^{-1} S_{\ell_{p,k}} = (\kappa_\infty + m_\Gamma(A - \ell_{p,k}))^{-1},$$

while two different rows retain the cross compliance

$$(10.29) \quad S_{\ell_i}^* K_{\Gamma}^{-1} S_{\ell_j}.$$

Thus all prime rows use one common regression and all off-diagonal phases survive. Equations (10.19) and (10.22) show that the matched finite-part identity is unchanged in the decharged frame.

10.6. The global polarization target. Let H_{CCM}^1 denote the separated resonant degree one supplied by the CCM cyclic construction, with weight-one scale action ϑ_t and descended alternating form Ω . The missing global object can now be stated independently of the finite matrix model.

Definition 10.4 (Global CCM polarization). A global polarization is a real-linear operator J on a dense scale-invariant core of H_{CCM}^1 such that

$$(10.30) \quad J^2 = -I, \quad J\vartheta_t = \vartheta_t J, \quad g(u, v) := \Omega(u, Jv)$$

extends to a faithful positive Hilbert metric. The normalized action $U_t = e^{-t/2}\vartheta_t$ must extend unitarily to its completion.

This is the infinite-dimensional analogue of the polarized degree-one page in the proof of the Riemann hypothesis for curves over finite fields, but the required positivity there is supplied by correspondences and the Hodge index theorem on a square, rather than by a formal Hilbert completion. Deninger's dynamical-cohomology program gives the same weight-one template [14].

Proposition 10.5 (Exact force of the polarization). *A global CCM polarization in the sense of Definition 10.4 implies RH.*

Proof. If a resonant class $u_\rho \neq 0$ has $\vartheta_t u_\rho = e^{t\rho} u_\rho$, then

$$U_t u_\rho = e^{t(\rho-1/2)} u_\rho.$$

Unitarity for every real t forces $e^{t(\Re\rho-1/2)} = 1$, hence $\Re\rho = 1/2$. The CCM resonant divisor is the nontrivial zero divisor of the completed zeta function, so all its points lie on the critical line. $\square$

Proposition 10.5 explains why positivity of the local pages cannot be promoted by a formal completion: the promotion itself contains the full missing sign.

10.7. The absolutely continuous completion no-go. The covariant system (10.20)–(10.29) has an ordinary Hilbert direct-limit completion. It is natural, positive and source-defined, but it has the wrong spectral type. The following operator-valued statement gives the precise obstruction.

Lemma 10.6 (Operator-valued dense-range criterion). *Let (X, μ) be sigma finite, let $\mathfrak{h}$ be separable, and let $M(x)$ be a measurable field of closed densely defined operators. If $\mathcal{M}$ is the associated maximal decomposable operator on $L^2(X, \mu; \mathfrak{h})$, then*

$$(10.31) \quad \boxed{\overline{\text{Ran } \mathcal{M}} = L^2(X, \mu; \mathfrak{h}) \iff \ker M(x)^* = \{0\} \text{ for almost every } x.}$$

If $M(x)^ M(x) \succeq \kappa I$ almost everywhere for one $\kappa > 0$, then $\mathcal{M}$ has closed range and is bounded below.*

Proof. The adjoint of the maximal decomposable operator has fibers $M(x)^*$. Therefore

$$(\overline{\text{Ran } \mathcal{M}})^\perp = \ker \mathcal{M}^* = \int_X^\oplus \ker M(x)^* d\mu(x),$$

which proves (10.31). The lower-bound assertion is the closed-range theorem applied fiberwise and then integrated. $\square$

Apply the lemma to the charged Gamma operator

$$(10.32) \quad K_{\Gamma, Q} = \kappa_\infty I + \bigoplus_{q \in Q} m_\Gamma(A - \log q).$$

It is uniformly bounded below by $\kappa_\infty I$. If the scalar term is omitted, the fiber kernels are supported on at most a countable set and the range is still dense. Unitary charge transports cannot change kernels of adjoints or closure of ranges.

Theorem 10.7 (Absolutely continuous completion no-go). *The cofinal Hilbert completion generated from the Cauchy coefficient module, shifted prime-power generators, Gamma graphs, covariant charge mixing and isometric direct limits has absolutely continuous normalized scale spectrum and no point spectrum. Consequently every equivariant linear map whose range vectors belong to this Hilbert target,*

$$(10.33) \quad D : H_{\text{CCM}}^1 \longrightarrow \mathcal{P}_\infty^{\text{cov}}$$

annihilates every resonant eigenclass. In particular it cannot be a faithful descent and cannot supply the polarization of Definition 10.4.

Proof. The measure in (10.20) is equivalent to Lebesgue measure. Countable orthogonal sums, shifts of the generator, unitary conjugacy, invariant closed subspaces, equivariant graph embeddings and isometric direct limits preserve absolute continuity. These are exactly the operations used in (10.20)–(10.29); hence the target has no eigenvectors.

For every critical-line zero $\rho = 1/2 + i\gamma$, the CCM degree one has a nonzero class satisfying $U_t^{\text{CCM}} u_\rho = e^{it\gamma} u_\rho$. Equivariance of (10.33) would make Du_ρ an eigenvector with the same character. Since the target has no point spectrum, $Du_\rho = 0$. Infinitely many such zeros exist unconditionally by Hardy's theorem [17]. $\square$

Remark 10.8 (Hilbert-target scope). Theorem 10.7 does not exclude an equivariant distributional map into the dual member Φ' of a rigged triple $\Phi \subset \mathcal{P}_\infty^{\text{cov}} \subset \Phi'$. Generalized eigenvectors of an absolutely continuous operator naturally live in such a dual. The theorem excludes faithful descent into the Hilbert member, which is exactly the target of the positive metric (10.27). A resonant nuclear construction must therefore retain evaluation vectors and jets in Φ' before asking which combinations admit a positive Hilbert completion.

The theorem proves a family-level statement with an explicit scope. It applies to decomposable absolutely continuous coefficient completions and to the operations listed in its proof. It does not assert that every conceivable Hilbert space is absolutely continuous. Its conclusion is the relevant dichotomy: retaining transverse charge fibers creates a spurious continuous cokernel, while removing them leaves dense range and zero reduced cokernel. Neither is the discrete resonant CCM degree one.

10.8. Six terminal formulations and their common content. The development reaches the same force-bearing sign through six distinct constructions. Their mathematical forms are different, but each becomes a proof of Weil positivity precisely when its final gate is supplied:

Realization	Terminal statement
Localized signed forms	Lower semicontinuity of the indefinite Weil form under the global exhaustion, not merely Mosco convergence of its positive shift.
Finite inertia/Schur analysis	The physical prime–Gamma gain exceeds the anti-shorter deficit, $\mathfrak{g}_{M,J,X} > \delta_{M,J,X}$, on a form-core exhaustion.
Fixed Rosati realization	A faithful factorization $\tau(f * g^\sharp) = \langle D[f], D[g] \rangle$; by GNS, its existence is equivalent to positivity of the Rosati/Weil form.
Alternative polarization	A source-defined positive Hilbert majorant for which normalized scaling is unitary and Ω is bounded and weakly nondegenerate.
Charged boundary pushout	Closure-faithful equivariant descent from the complete Tate–Gamma–polar source to the separated CCM cokernel.
Geometric degree one	A positive global polarization $(H_{\text{CCM}}^1, \Omega, J, g)$ as in Definition 10.4.

The first two rows localize the sign in finite matrices. The third states the sign as a factorization. The fourth and fifth try to construct a new positive metric before comparison with Rosati. The last states the desired geometric structure directly. None of the implications from its terminal statement to RH is missing; the missing assertion in each row is the same global sign in the corresponding category.

This repeated endpoint admits a general formulation.

Proposition 10.9 (Deformation-blindness obstruction). *Let $\mathcal{X}$ be a class of divisor-bearing objects, let $\sim$ be a deformation relation on $\mathcal{X}$, and let $\kappa : \mathcal{X} \rightarrow \mathbb{Z}$ be an index depending on the position of the divisor. Suppose that a functor $F : \mathcal{X} \rightarrow \mathcal{Y}$ is constant on $\sim$ -classes. If κ is not constant on one such class, then no map $\phi : F(\mathcal{X}) \rightarrow \mathbb{Z}$ satisfies $\kappa = \phi \circ F$.*

Proof. Choose $x \sim y$ with $\kappa(x) \neq \kappa(y)$. Then $F(x) = F(y)$, whereas a factorization would give $\kappa(x) = \phi(F(x)) = \phi(F(y)) = \kappa(y)$, a contradiction. $\square$

In the present program, Proposition 10.9 organizes four otherwise separate obstructions. The table states the forgetful functor actually used, the relation through which it factors, and an explicit witness showing that the required index does not factor through it.

Functor F	Relation R forgotten by F	Witness with different index information
Half-plane sign class of the Maass–Selberg Cauchy pairing	Motion of resonance parameters inside $0 < \Re \rho < 1$	Writing $w_\rho = (1 - \rho)/2$, the kernel is always $-1/(w_\rho + \overline{w_\sigma})$, hence negative definite. Five on-line resonances give signature $(0, 5)$; adjoining an off-line quartet gives $(0, 9)$, still definite, although the Weil off-line index changes.
Scalar determinant section	Equality of determinants after forgetting the sourced Hermitian kernel and its Krein–Langer divisor	The Hermitian matrices I_2 and $-I_2$ both have determinant one, but negative indices zero and two. The same ambiguity is why a Nevanlinna determinant must be augmented by its sourced second variation and divisor grading.
Cyclic Chern character	Bounded operator homotopy, including scalar shifts of a finite semilocal Weil operator	The path from an unshifted finite Weil matrix to its lower-edge-shifted nonnegative matrix preserves the character, while the lower edge and the negative eigenvalue count can change.
Curvature of a positive determinant-line metric	Motion of a section divisor with the Hermitian line and its metric fixed	On a disk, $s_a(z) = z - a$ and $s_b(z) = z - b$ have different divisors, but the same curvature $\partial\bar{\partial}\phi$ for $\ 1\ ^2 = e^{-\phi}$. Poincaré–Lelong records the moved divisor but metric positivity does not localize it.

Theorem 10.7 is deliberately not a fifth row of this table. Its proof does not require a deformation witness. It is a spectral-type obstruction: the source has point-spectrum eigenclasses, the Hilbert target has purely absolutely continuous spectrum, and equivariance forces every such class to map to zero.

The conclusion is not that every analytic limit is signature-blind. Spectral flow, for example, can retain an index when a gap and a continuous Fredholm path are supplied. The conclusion is narrower and operational: none of the four functors in the table can recover the off-line index because it factors through an equivalence relation under which that index can change; and the absolutely continuous Hilbert completion fails for the independent point-versus-continuous spectral reason.

10.9. Two open programs. The preceding construction and no-go leave two mathematically distinct directions. Both must be divisor-sensitive from the start; neither may define its positive metric from the zeros or from the Weil form whose sign is sought.

10.9.1. A geometric intersection product and Hodge-index theorem. The first direction is to construct the missing geometric source of positivity. The finite identity (10.7) and the archimedean identity (10.12) specify its local intersection numbers and determinant fibers. What is still absent is a global square carrying correspondences, a relative intersection form, and a primitive Hodge-index theorem. A successful object must provide:

- (1) classes Γ_{p^k} whose local torsion and incidence recover $\Lambda(p^k)/\sqrt{p^k}$;
- (2) an archimedean fiber whose determinant is (10.12), including the polar H^0/H^2 page;
- (3) a Lefschetz or supertrace formula identifying the resulting intersection distribution with Weil’s explicit formula;
- (4) a primitive intersection inequality whose induced Rosati form on degree one is positive;

- (5) compatibility with the scaling correspondences, so that normalized scaling is unitary on the polarized degree one.

This is the analogue of the role played by correspondences, the Jacobian and the Hodge index theorem in Weil's proof over finite fields. The work of Connes and Consani on the arithmetic curve, its Jacobian and absolute geometry supplies the natural ambient program [3, 7, 8]. Theorems 10.1 and 10.2 give concrete local constraints that any such global theory must satisfy. They do not supply the global intersection inequality.

10.9.2. *A divisor-sensitive resonant completion.* The second direction remains analytic or derived. It must replace the ordinary direct-integral completion by a topology in which evaluation and multiplicity jets at a discrete divisor survive before Hilbert reduction. At minimum one needs a source-defined space $\mathcal{E}$, a closed restriction complex and a map

$$(10.34) \quad H_{\text{CCM}}^1 \longrightarrow H^1(\mathcal{E})$$

with the following properties:

- (1) (10.34) is faithful on every resonant jet and is equivariant for scaling;
- (2) the Tate, Gamma and polar boundary maps act continuously and retain the exact finite part (10.19);
- (3) the alternating residue form extends boundedly and nondegenerately;
- (4) a positive compatible complex structure is constructed from source data, not from the zero divisor or the positive part of the Weil form;
- (5) the resulting normalized scale action is unitary.

Possible ambient categories include nuclear analytic spaces, derived cyclic modules and reproducing-kernel completions. The category alone is not a solution: the classical de Branges program shows that bounded point evaluation can simply re-express the missing Hermite–Biehler sign, and specific proposed positivity conditions for zeta and L -functions are known to fail [15, 16]. The acceptance test is therefore exact: the construction must fail for a Davenport–Heilbronn type function without an Euler product [18], must preserve the CCM divisor, and must obtain positivity from arithmetic source structure rather than assume it.

The two programs are related but not identical. A geometric intersection theory could produce the resonant completion as its analytic realization. Conversely, a successful divisor-sensitive completion might reveal the one-dimensional shadow of the missing intersection theory. At present both positivity theorems remain open.

11. CONCLUSION

The Connes proof of RH, in its current mathematical form, has the following shape:

$$(11.1) \quad \begin{array}{l} \text{finite semilocal Weil self-adjointness} \\ + \text{finite critical-line prolate zeta-cycles} \\ + \text{Suzuki localized operator realization} \\ + \text{a faithful positive global polarization} \\ \implies \mathcal{Q}_W \geq 0 \implies \text{RH.} \end{array}$$

The first three blocks are supplied by the published CCM/Connes/Suzuki programs in the precise sense stated in §§2–7. The finite analysis of §§8–9 proves that approximation, tail control, radical shorting, prime sampling, adaptive gain recovery and form-core exhaustion create no additional

sign loss. In that realization the terminal assertion is the physical surplus $\mathfrak{g}_{M,J,X} > \delta_{M,J,X}$ on the heat or hybrid rows.

Section 10 then reconstructs the source geometry behind the same assertion. At every finite prime, a positive cyclic-root Laplacian produces the relative torsion $\log p$ and the incidence $p^{-k/2}$, hence the exact coefficient $\Lambda(p^k)/\sqrt{p^k}$. At infinity, the positive Gamma spin and the polar plane reproduce the completed archimedean determinant. The common boundary has an exact matched finite part, and the charge translations assemble all prime powers equivariantly without removing cross phases. These statements are unconditional and source-defined.

They do not complete (11.1). Theorem 10.7 proves that the ordinary absolutely continuous Hilbert completion of this source object has no point spectrum and therefore cannot receive the resonant CCM degree one faithfully. Thus the finite and local polarizations are valid, while the attempted global descent is not. This separates a successful local construction from the still missing global Hodge/Rosati positivity.

The six terminal formulations in §10.8 are not six independent missing lemmas. They are six categorical expressions of the complete Weil sign. Proposition 10.9 identifies their recurring deformation failure: scalar determinants, homotopy classes and curvature factor through relations which forget the position or signature of the resonant divisor. The absolutely continuous Hilbert pushout fails for the independent spectral reason of Theorem 10.7. Together these results explain why further repackaging by the same operations cannot create the missing polarization.

Two routes remain open. The geometric route asks for an arithmetic intersection product and a primitive Hodge-index theorem whose local fibers are constrained by Theorems 10.1 and 10.2. The divisor-sensitive route asks for a resonant nuclear or derived completion which retains every evaluation jet before Hilbert reduction and constructs a positive scale-compatible metric from arithmetic source data. Either successful construction would supply the fourth block of (11.1). Neither construction, nor RH, is proved here.

REFERENCES

- [1] A. Weil, *Sur les “formules explicites” de la theorie des nombres premiers*, Comm. Sem. Math. Univ. Lund (1952), 252–265.
- [2] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.
- [3] A. Connes and C. Consani, *Weil positivity and trace formula, the archimedean place*, arXiv:2006.13771 (2020).
- [4] A. Connes and W. D. van Suijlekom, *Quadratic forms, real zeros and echoes of the spectral action*, Comm. Math. Phys. **406** (2025), no. 12, Paper No. 312.
- [5] A. Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022 (2026).
- [6] A. Connes, C. Consani, and H. Moscovici, *Zeta spectral triples*, arXiv:2511.22755 (2025).
- [7] A. Connes and C. Consani, *On the Jacobian of $\overline{\text{Spec}} \mathbb{Z}$* , arXiv:2602.15941 (2026).
- [8] A. Connes and C. Consani, *On the Absolute Geometry of $\text{Spec} \mathbb{Z}$* , arXiv:2606.06604 (2026).
- [9] M. Suzuki, *On the Hilbert space derived from the Weil distribution*, Canadian Journal of Mathematics, FirstView (2025).
- [10] M. Suzuki, *Detecting zeros of Dirichlet L-functions via the Riemann zeta-function*, arXiv:2508.17701 (2025).
- [11] M. Suzuki, *Weil’s quadratic form via the screw function*, arXiv:2606.09096 (2026).
- [12] J.-F. Burnol, *Two complete and minimal systems associated with the zeros of the Riemann zeta function*, J. Théor. Nombres Bordeaux **16** (2004), no. 1, 65–94.
- [13] J.-F. Burnol, *Entrelacement de co-Poisson*, Ann. Inst. Fourier (Grenoble) **57** (2007), no. 2, 525–602.
- [14] C. Deninger, *Some analogies between number theory and dynamical systems on foliated spaces*, Doc. Math., Extra Volume ICM 1998, vol. I, 163–186.
- [15] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice–Hall, Englewood Cliffs, NJ, 1968.
- [16] J. B. Conrey and X.-J. Li, *A note on some positivity conditions related to zeta and L-functions*, Internat. Math. Res. Notices (2000), no. 18, 929–940.
- [17] G. H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, C. R. Acad. Sci. Paris **158** (1914), 1012–1014.

- [18] H. Davenport and H. Heilbronn, *On the zeros of certain Dirichlet series*, J. London Math. Soc. **11** (1936), 181–185, 307–312.
- [19] T. M. Dunster, *Asymptotics of prolate spheroidal wave functions*, arXiv:1601.00699v3 (2017).
- [20] M. G. Krein, *The theory of self-adjoint extensions of semi-bounded Hermitian transformations and its applications. I*, Mat. Sb. (N.S.) **20(62)** (1947), 431–495.
- [21] W. N. Anderson, Jr., *Shorted operators*, SIAM J. Appl. Math. **20** (1971), no. 4, 520–525.
- [22] W. N. Anderson, Jr. and G. E. Trapp, *Shorted operators. II*, SIAM J. Appl. Math. **28** (1975), no. 1, 60–71.
- [23] E. V. Haynsworth, *Determination of the inertia of a partitioned Hermitian matrix*, Linear Algebra Appl. **1** (1968), 73–81.
- [24] F. R. Gantmacher, *The Theory of Matrices*, Chelsea, New York, 1959.
- [25] K. Loewner, *Über monotone Matrixfunktionen*, Math. Z. **38** (1934), 177–216.
- [26] R. Bhatia, *Matrix Analysis*, Graduate Texts in Mathematics, vol. 169, Springer, 1997.
- [27] S. Karlin, *Total Positivity, Vol. I*, Stanford University Press, 1968.
- [28] M. E. H. Ismail, *Classical and Quantum Orthogonal Polynomials in One Variable*, Encyclopedia of Mathematics and its Applications, vol. 98, Cambridge University Press, 2005.
- [29] U. Mosco, *Convergence of convex sets and of solutions of variational inequalities*, Advances in Math. **3** (1969), 510–585.
- [30] K. Kuwae and T. Shioya, *Convergence of spectral structures: a functional analytic theory and its applications to spectral geometry*, Comm. Anal. Geom. **11** (2003), no. 4, 599–673.
- [31] O. Post, *Spectral Analysis on Graph-like Spaces*, Lecture Notes in Mathematics, vol. 2039, Springer, 2012.
- [32] K. Suzuki and T. Uemura, *On instability of global path properties of symmetric Dirichlet forms under Mosco-convergence*, arXiv:1412.0725 (2014).